

Pellicer Creek Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

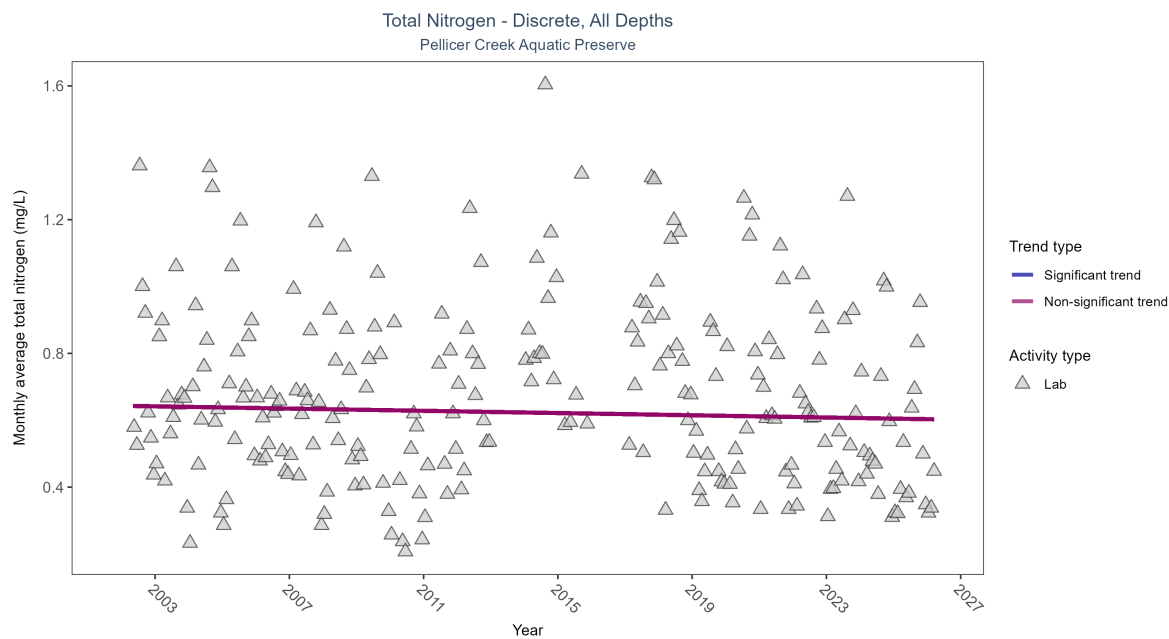


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1991	23	2002 - 2026	0.6366	-0.03735	0.64324	-0.00167	0.4279

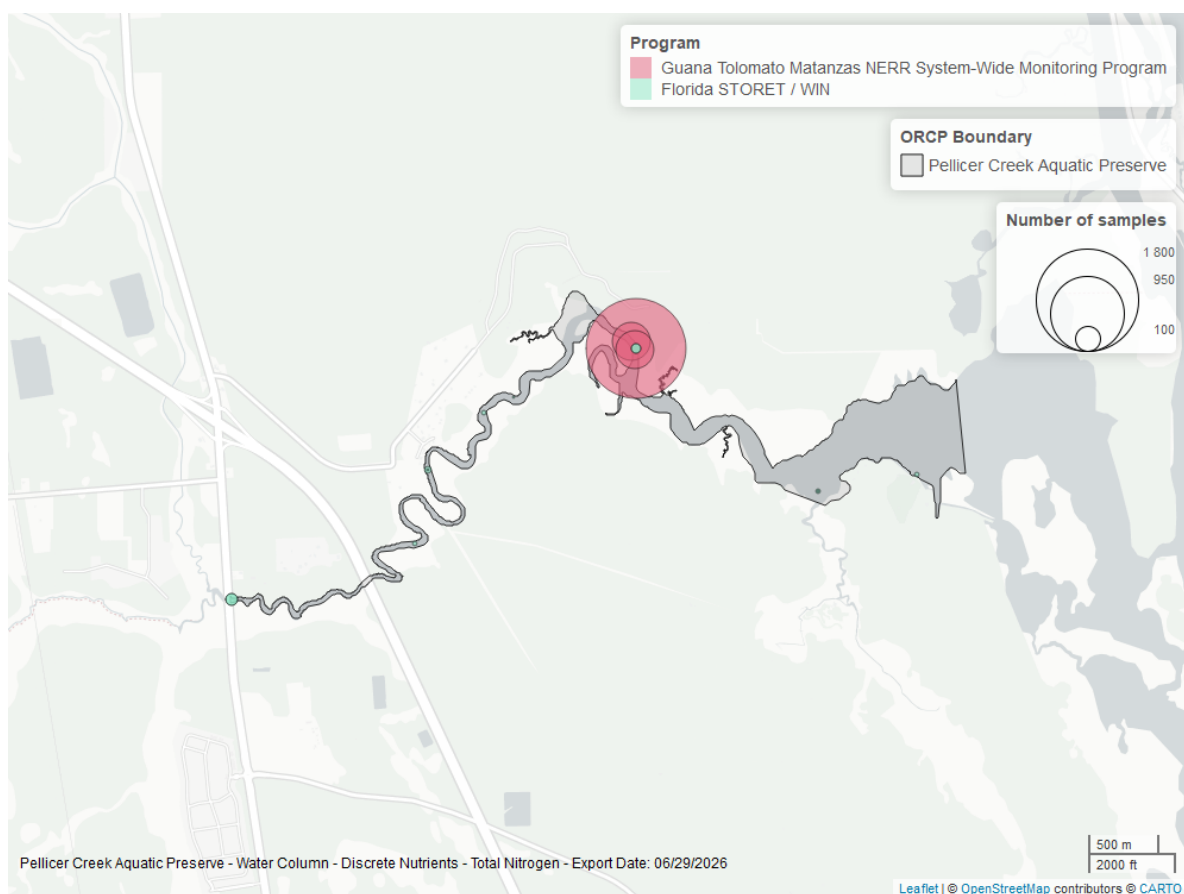


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

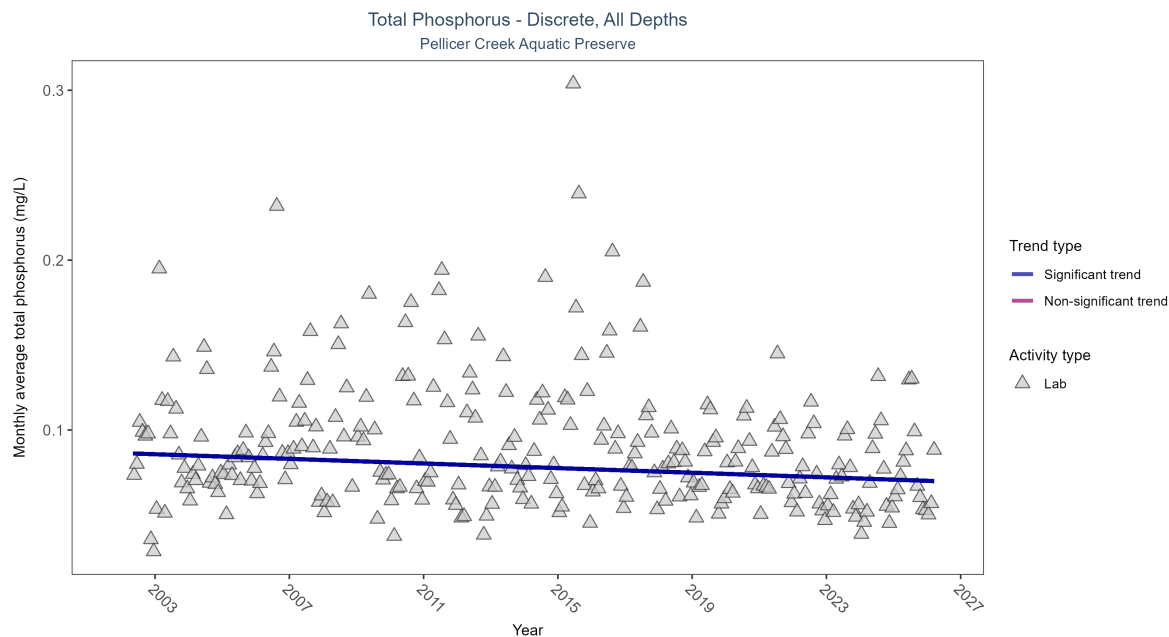


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	3815	25	2002 - 2026	0.08	-0.18911	0.08643	-0.00068	0

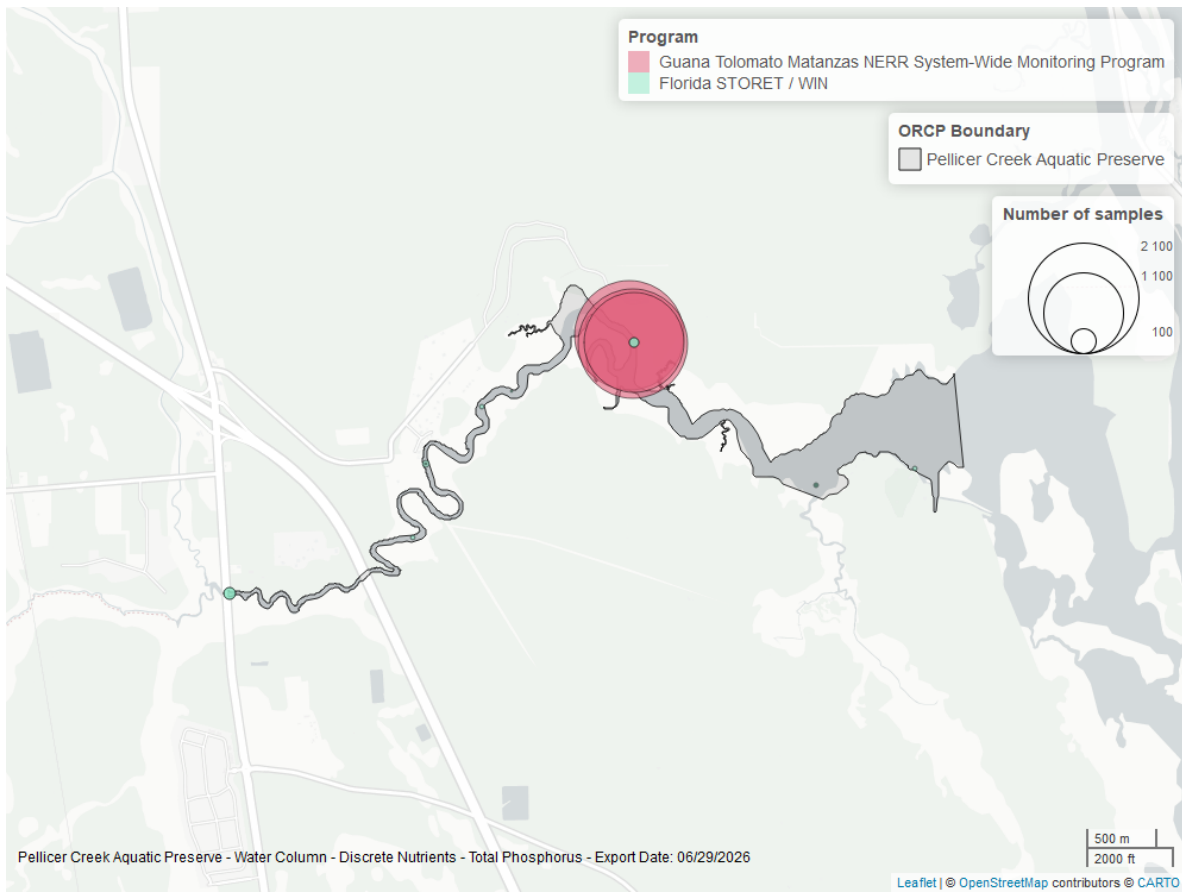


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

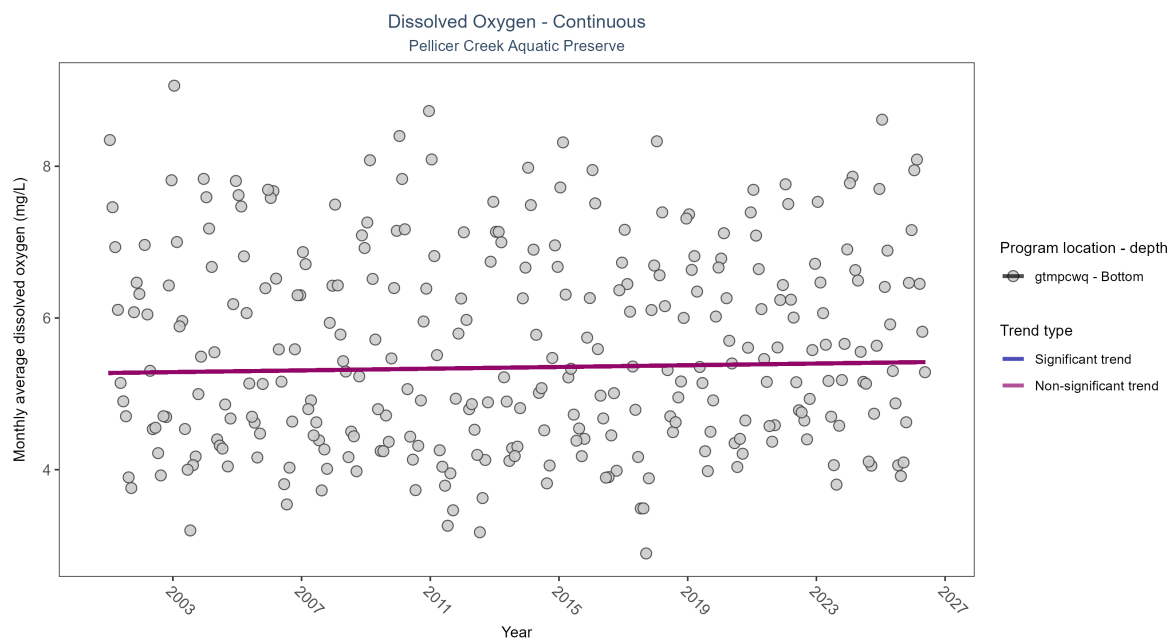


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: No detectable change in monthly average dissolved oxygen was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpcwq	No detectable trend	746219	26	2001 - 2026	5.5	0.05	5.27	0.01	0.17

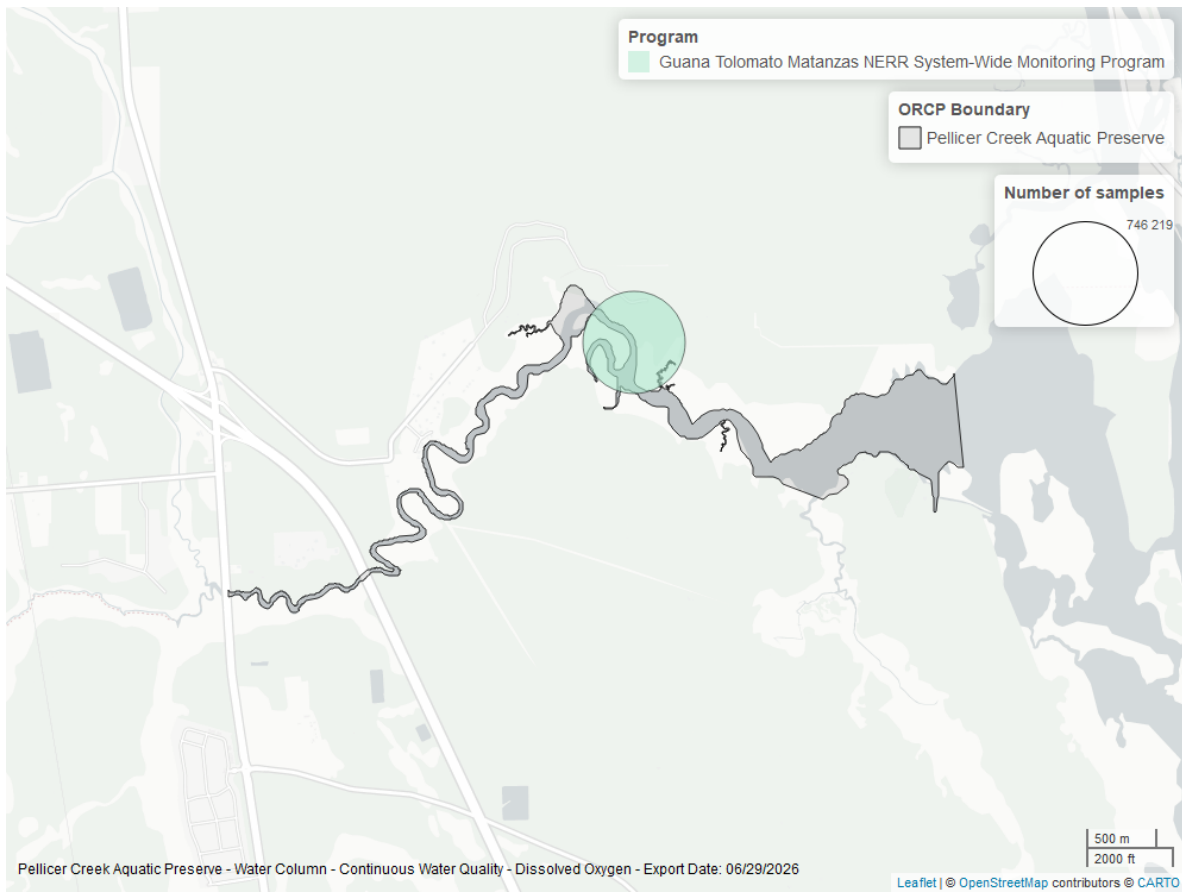


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

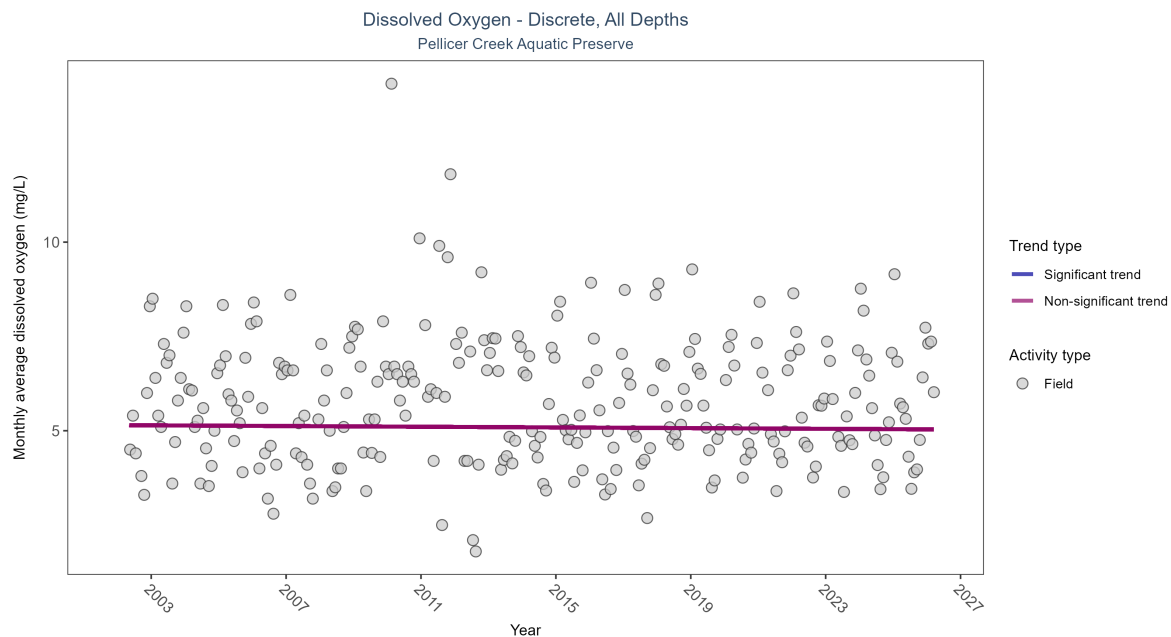


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	2320	25	2002 - 2026	5.645	-0.01968	5.14738	-0.00462	0.6529

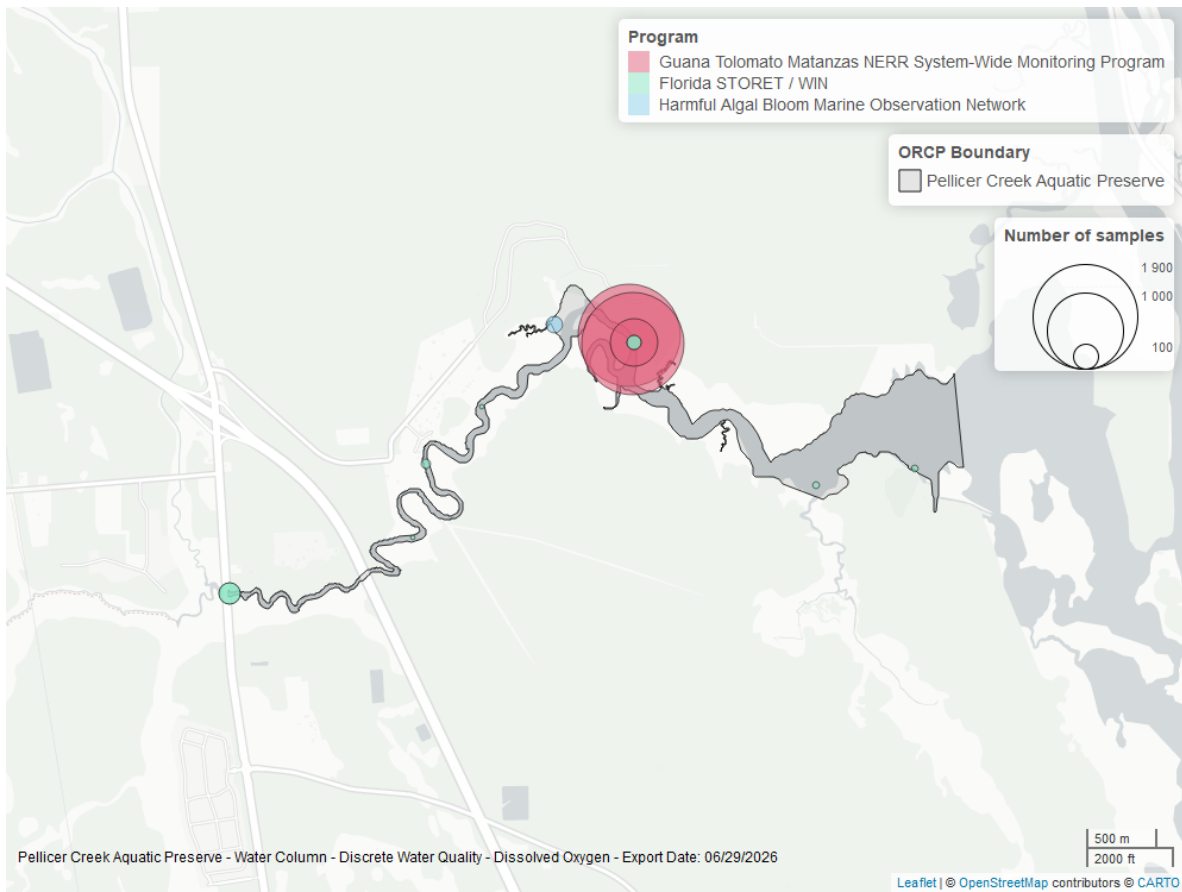


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

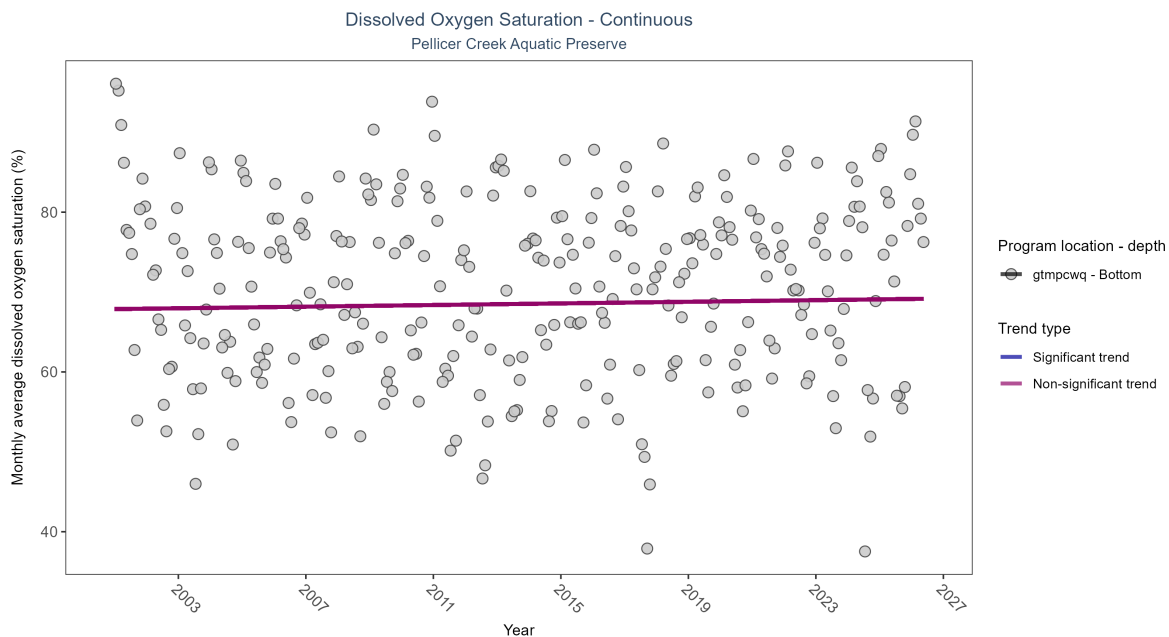


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: No detectable change in monthly average dissolved oxygen saturation was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpcwq	No detectable trend	747712	26	2001 - 2026	71.7	0.04	67.87	0.05	0.35

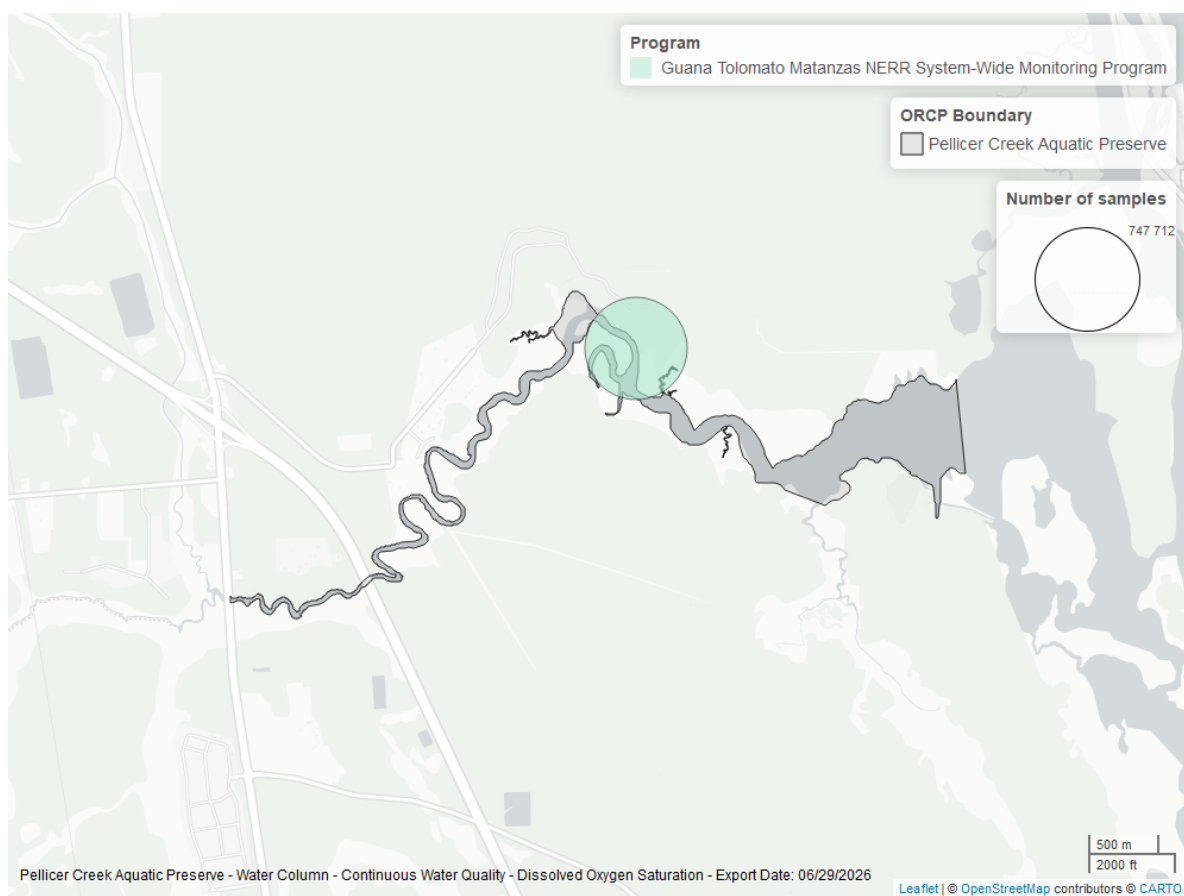


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

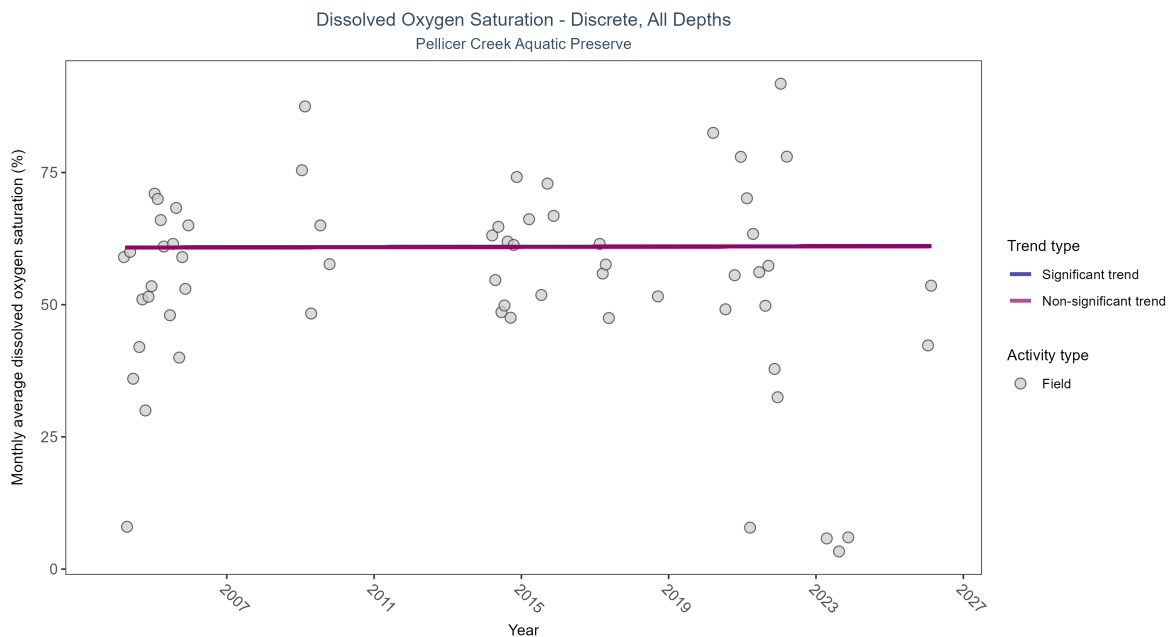


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Dissolved oxygen saturation showed no detectable trend between 2004 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	203	12	2004 - 2026	55.4	-0.00092	60.80703	0.0125	0.9508

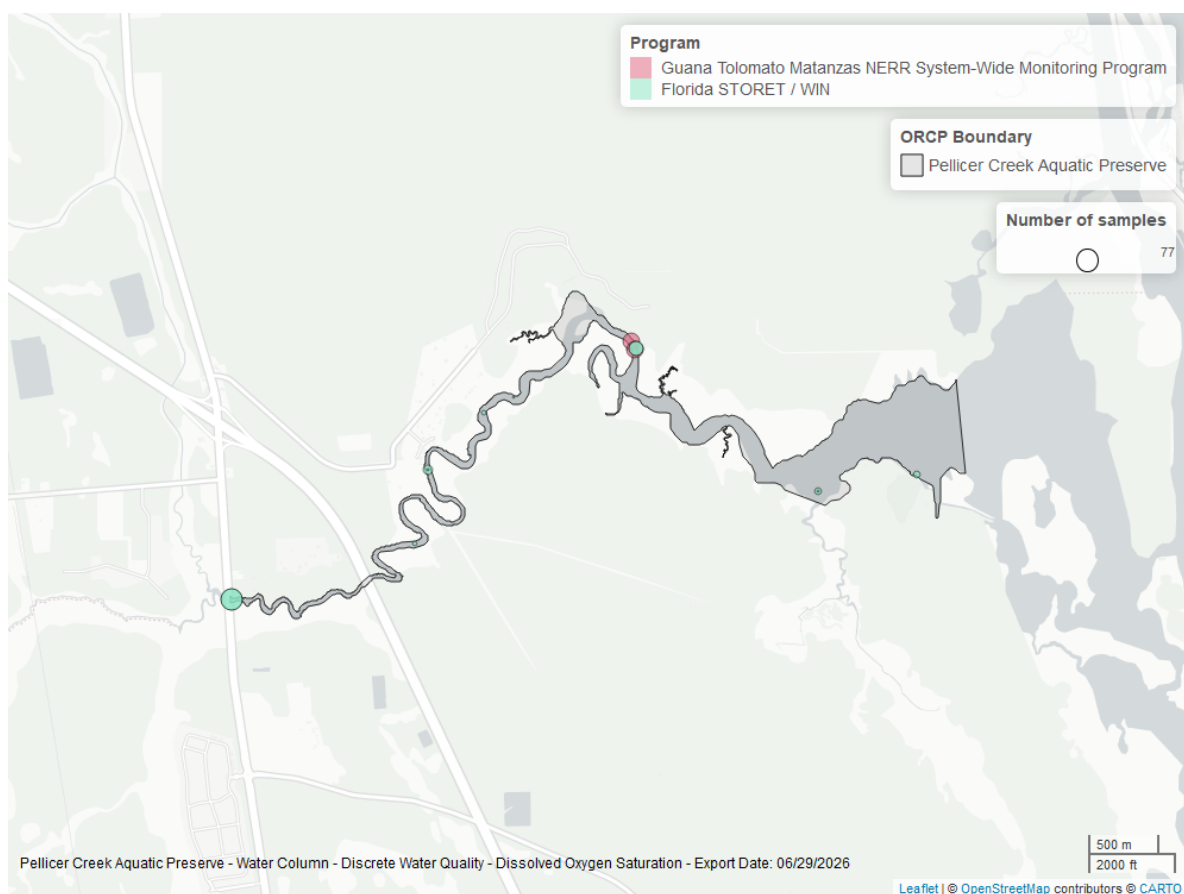


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

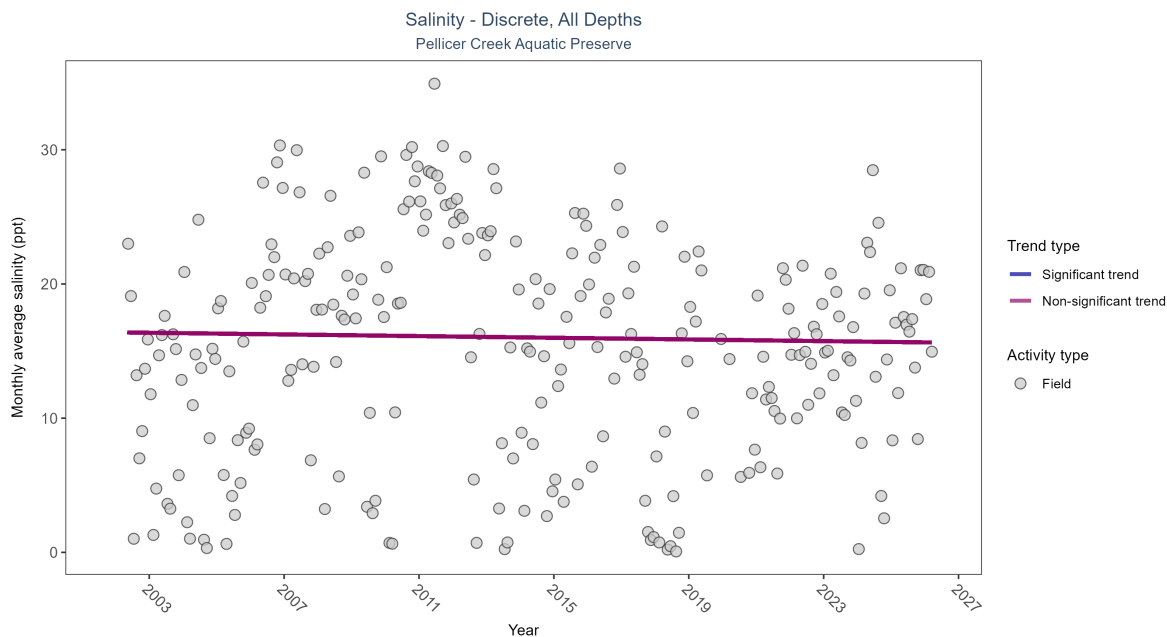


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Salinity showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	3130	25	2002 - 2026	16.335	-0.02277	16.39596	-0.0312	0.6281

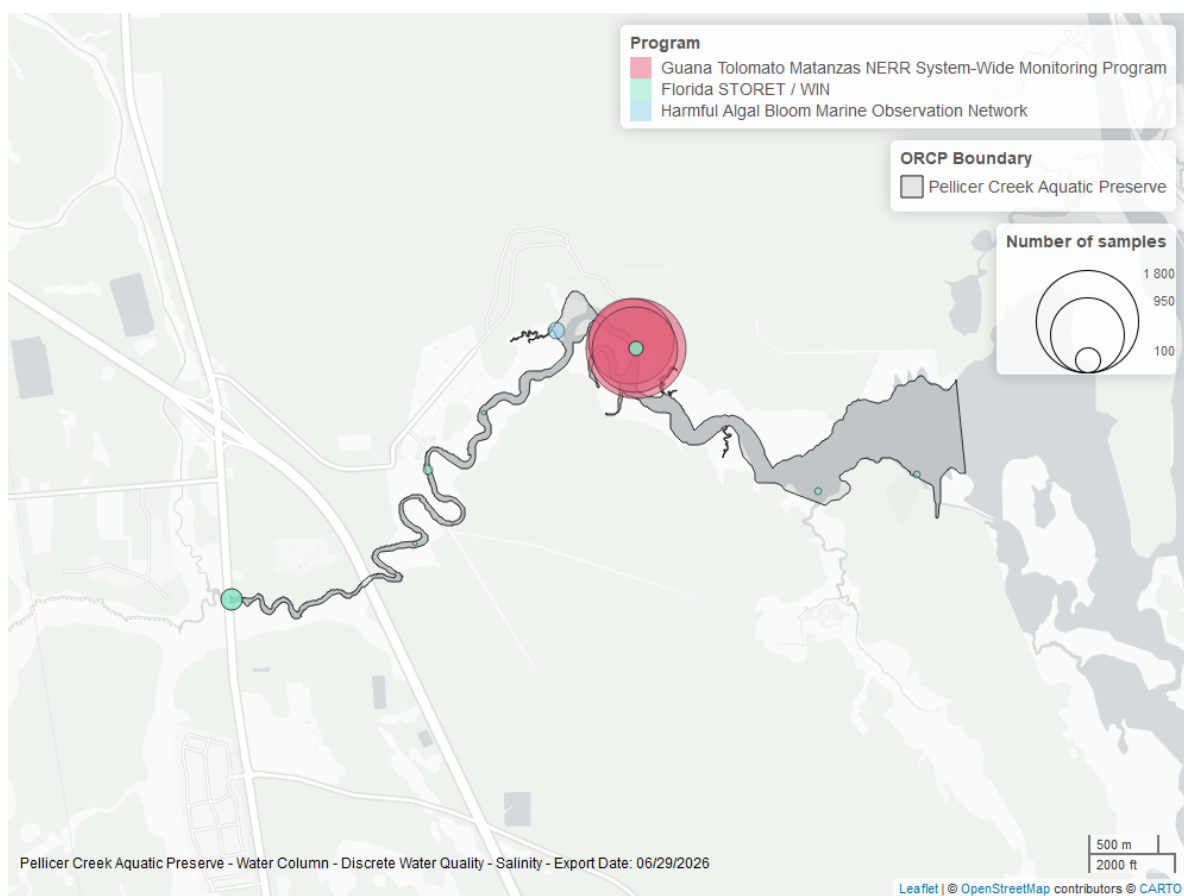


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

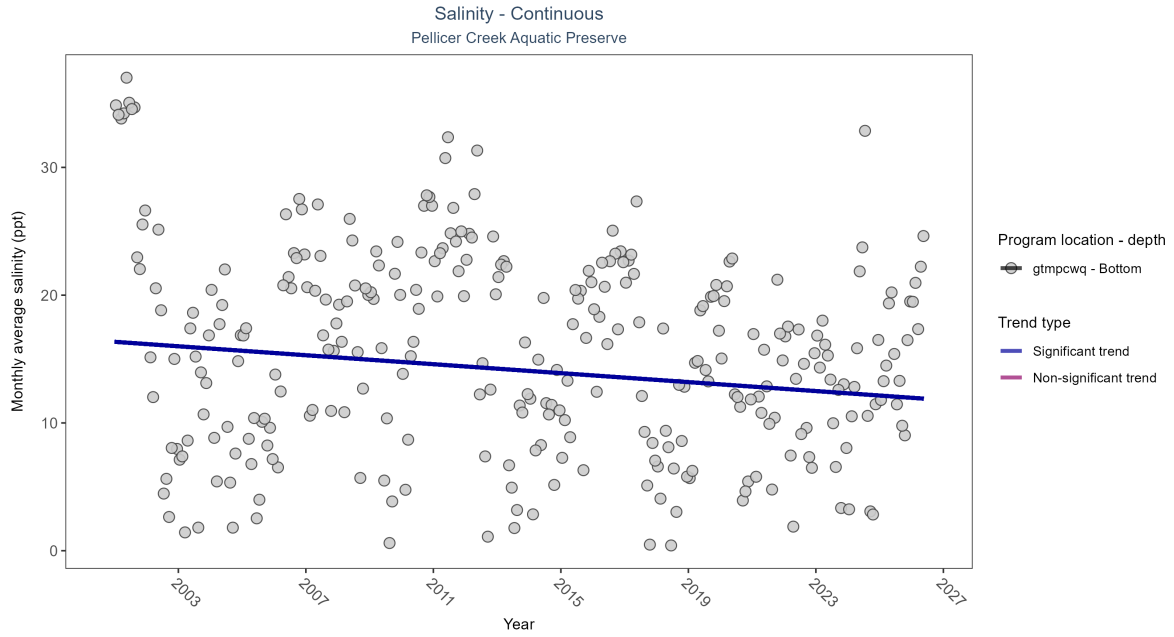


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At one program location, monthly average salinity decreased by 0.18 ppt per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpcwq	Decreasing trend	755999	26	2001 - 2026	16.7	-0.12	16.35	-0.18	0.01

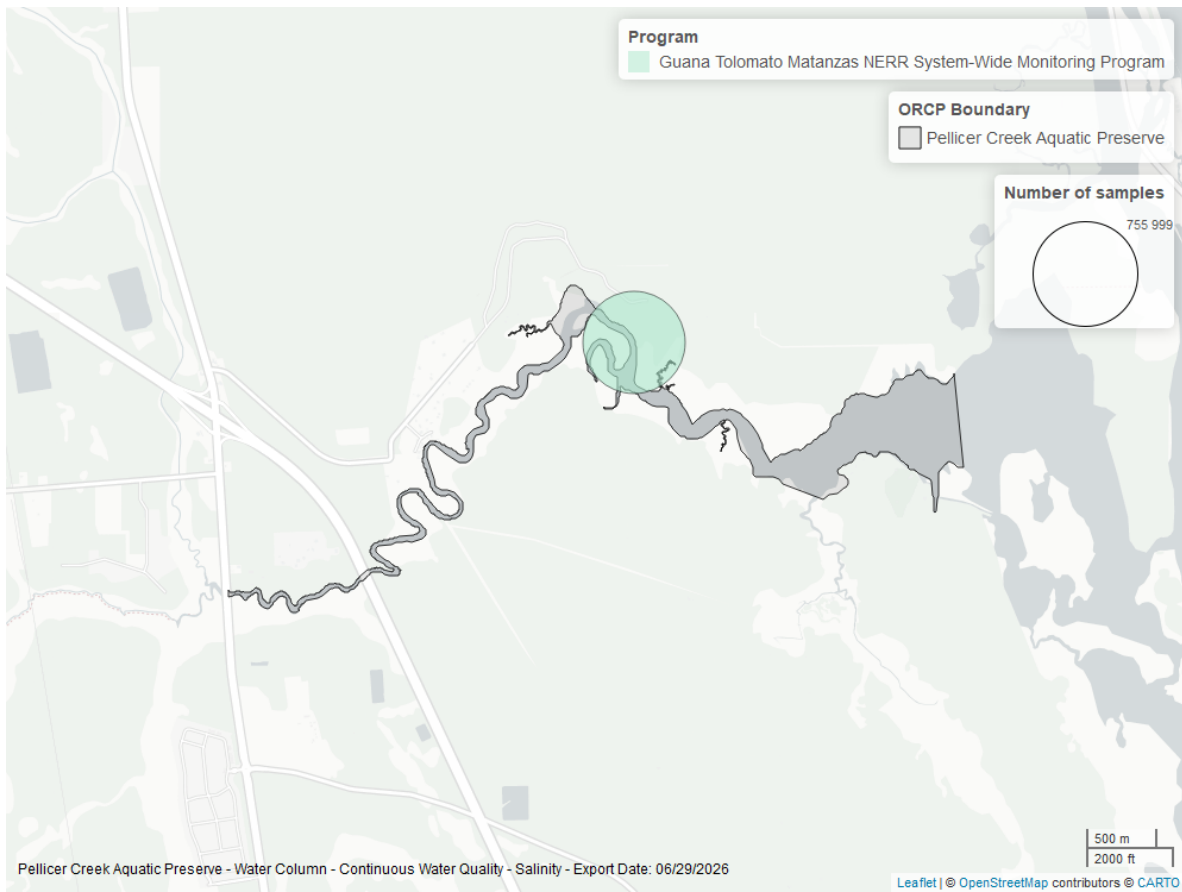


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

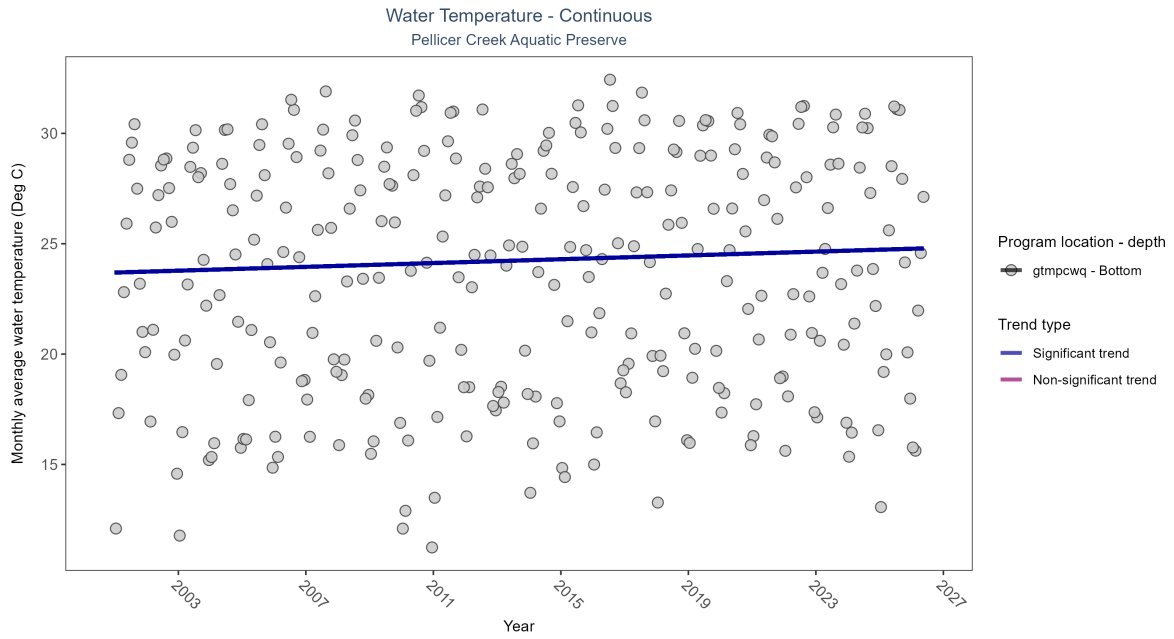


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At one program location, monthly average water temperature increased by 0.04Deg C per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpcwq	Increasing trend	764565	26	2001 - 2026	24.3	0.16	23.69	0.04	0

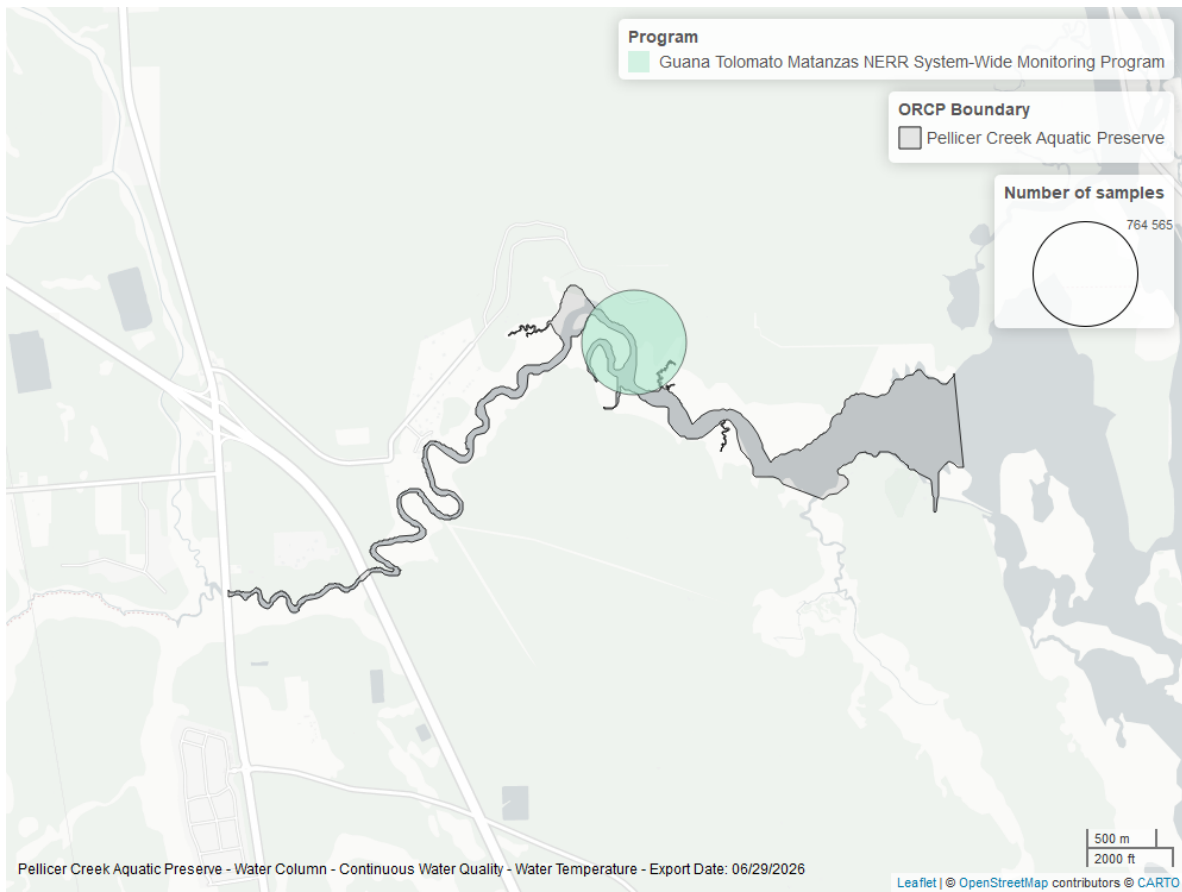


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

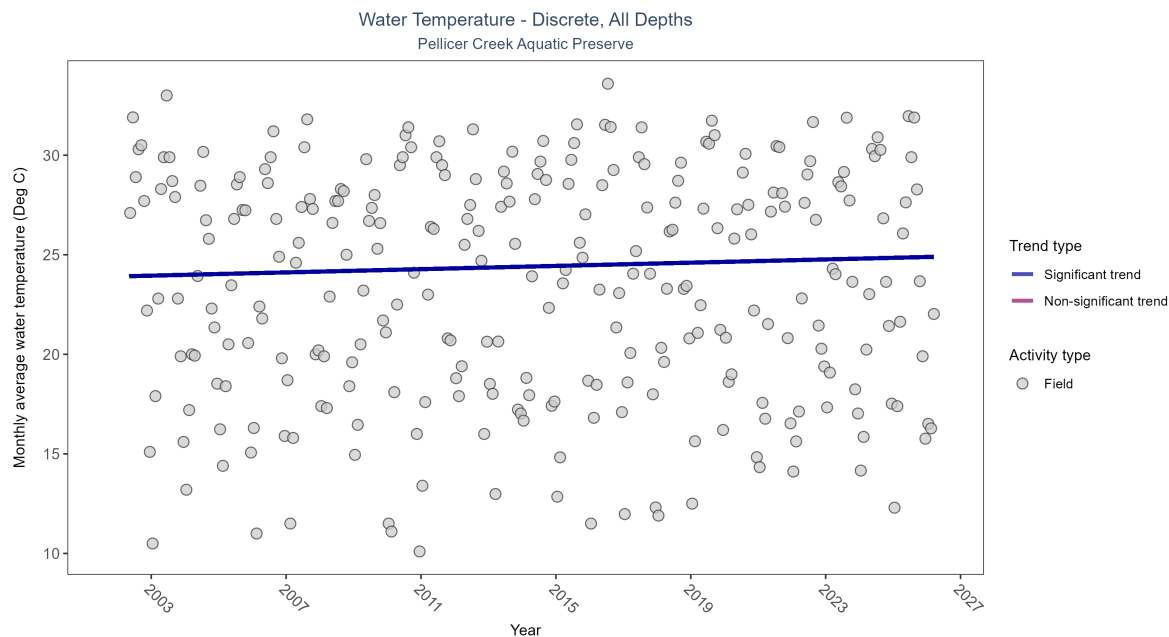


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Monthly average water temperature increased by 0.04Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	2474	25	2002 - 2026	23.727	0.08816	23.9098	0.04074	0.0382

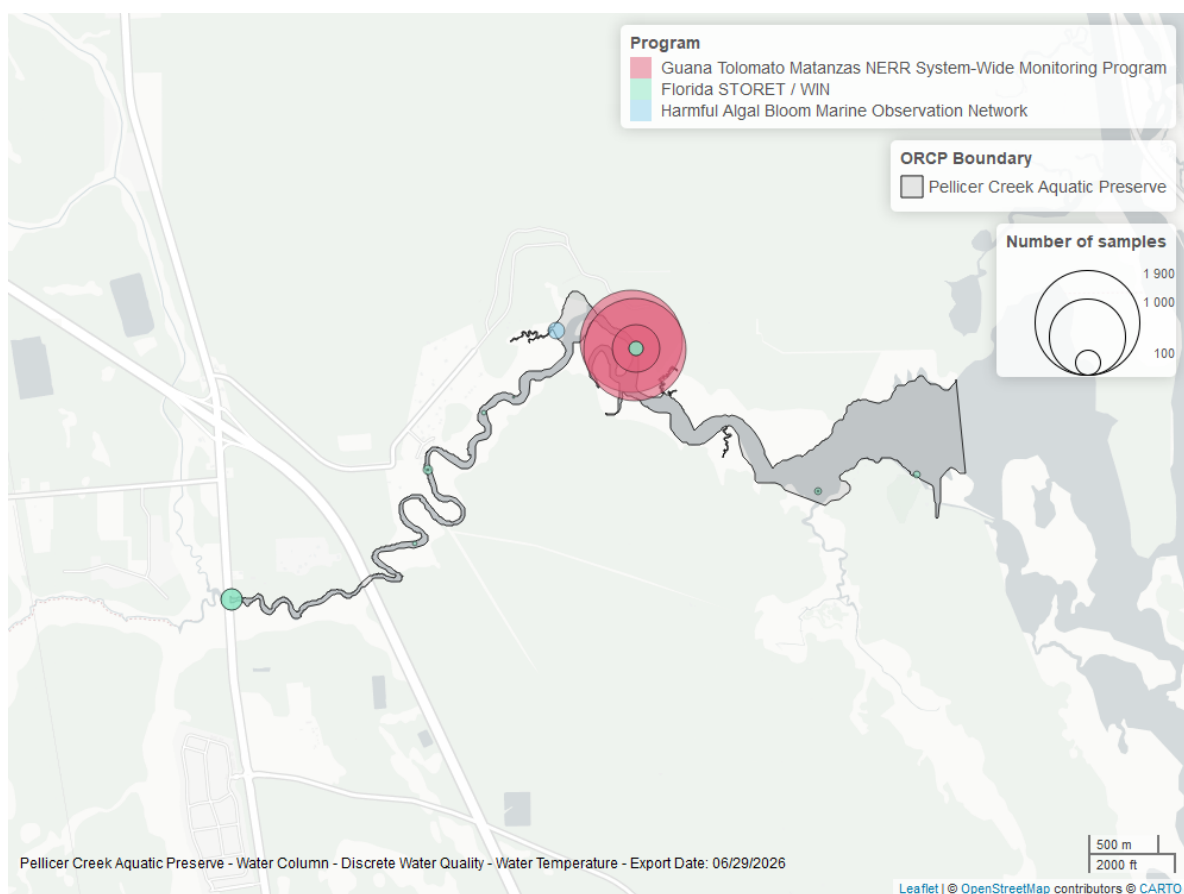


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

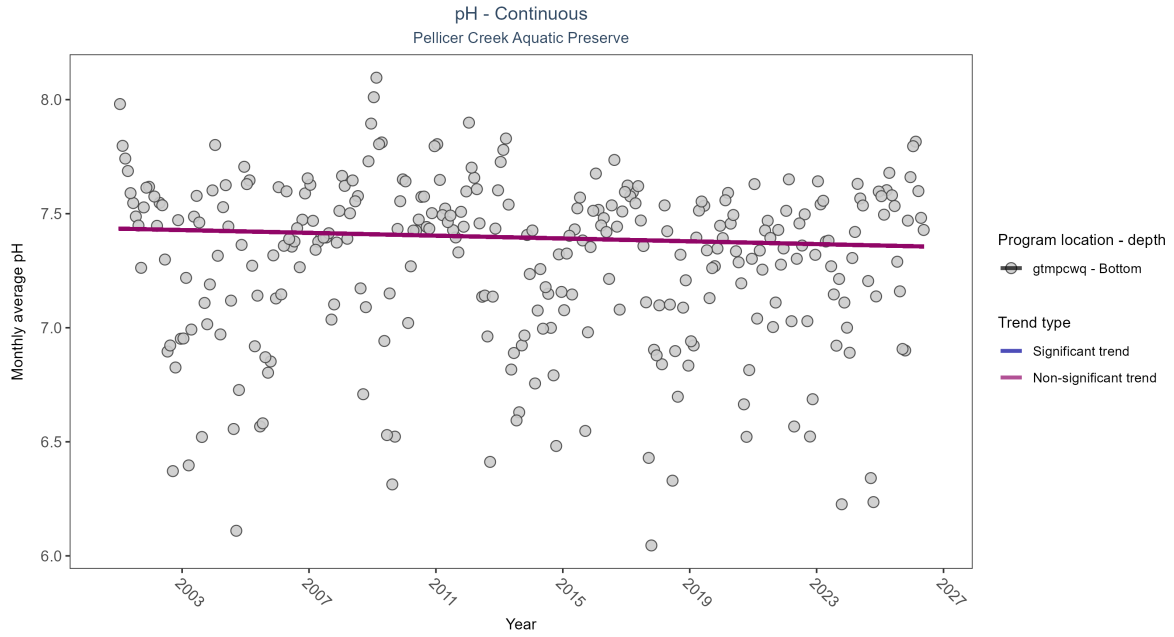


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: No detectable change in monthly average pH was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpcwq	No detectable trend	740446	26	2001 - 2026	7.4	-0.07	7.43	0	0.07

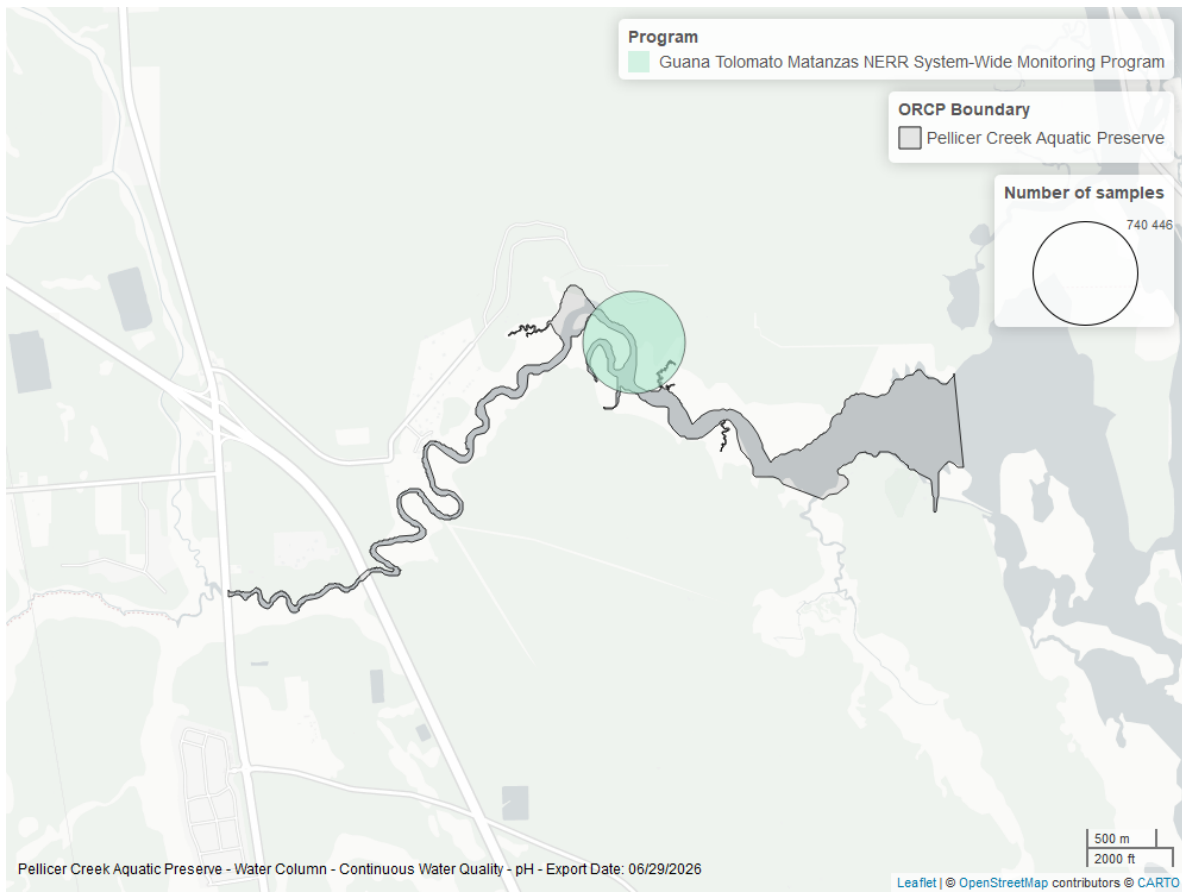


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

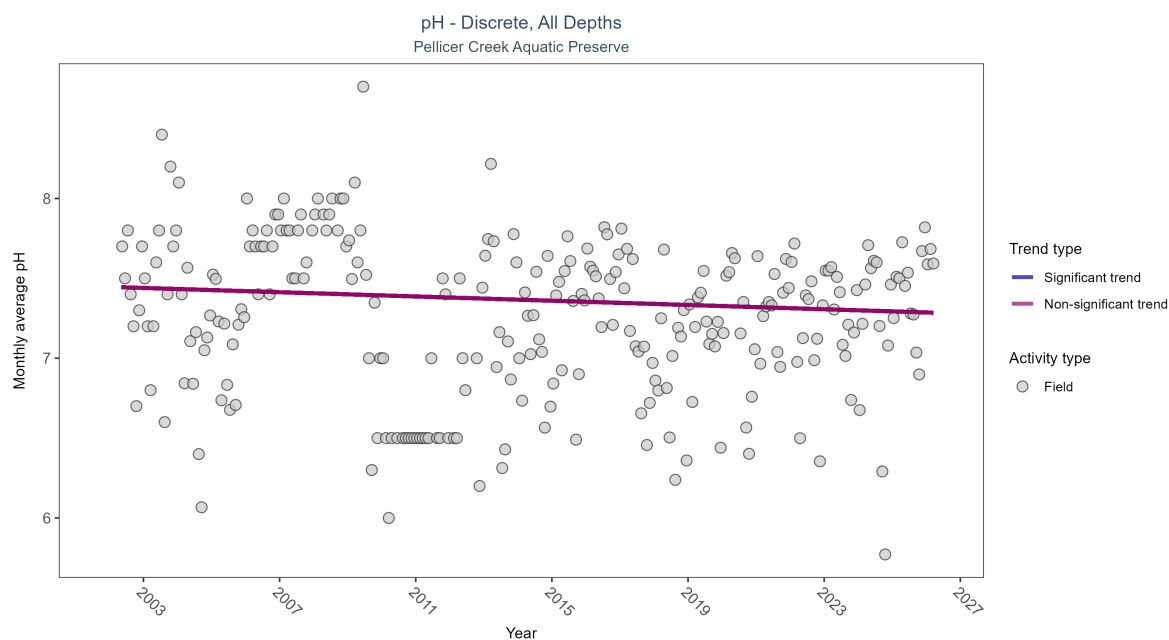


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: pH showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	2382	25	2002 - 2026	7.3	-0.06737	7.44672	-0.00667	0.1185

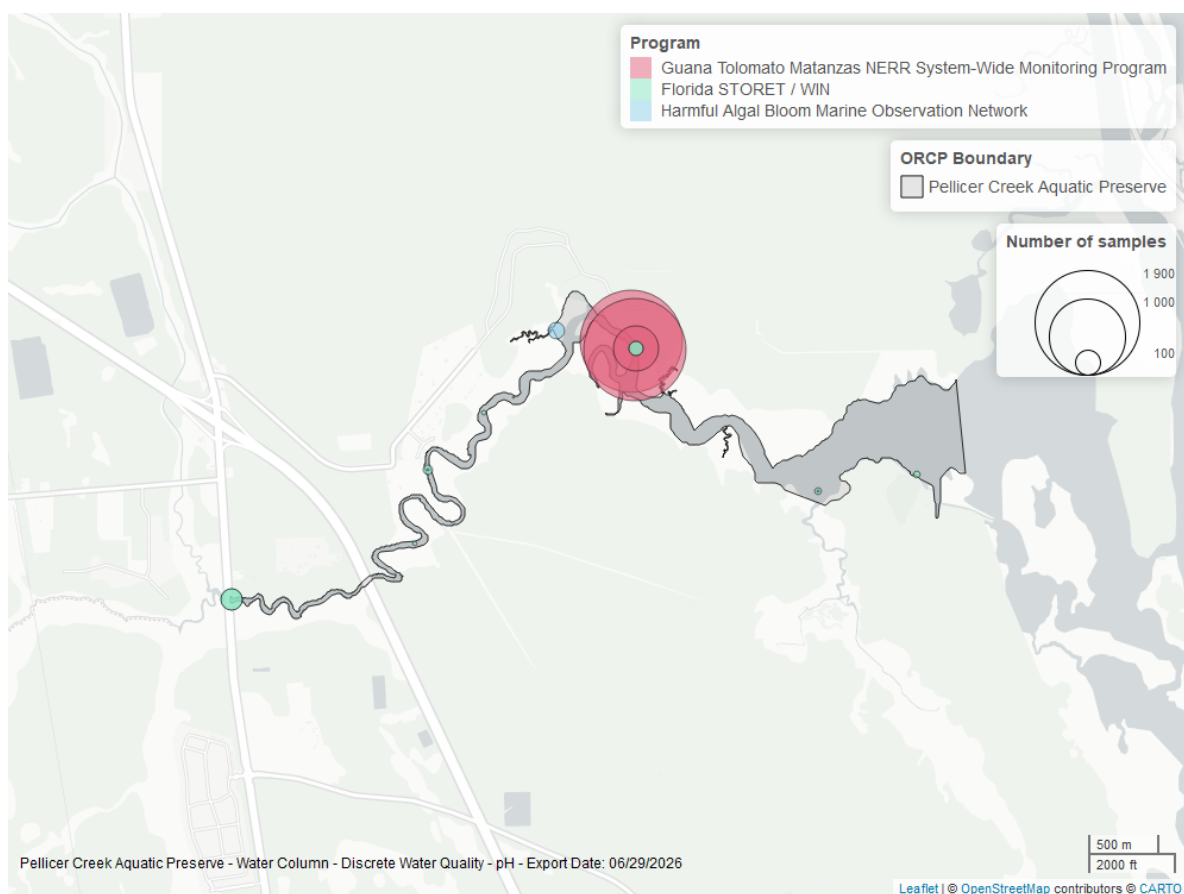


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

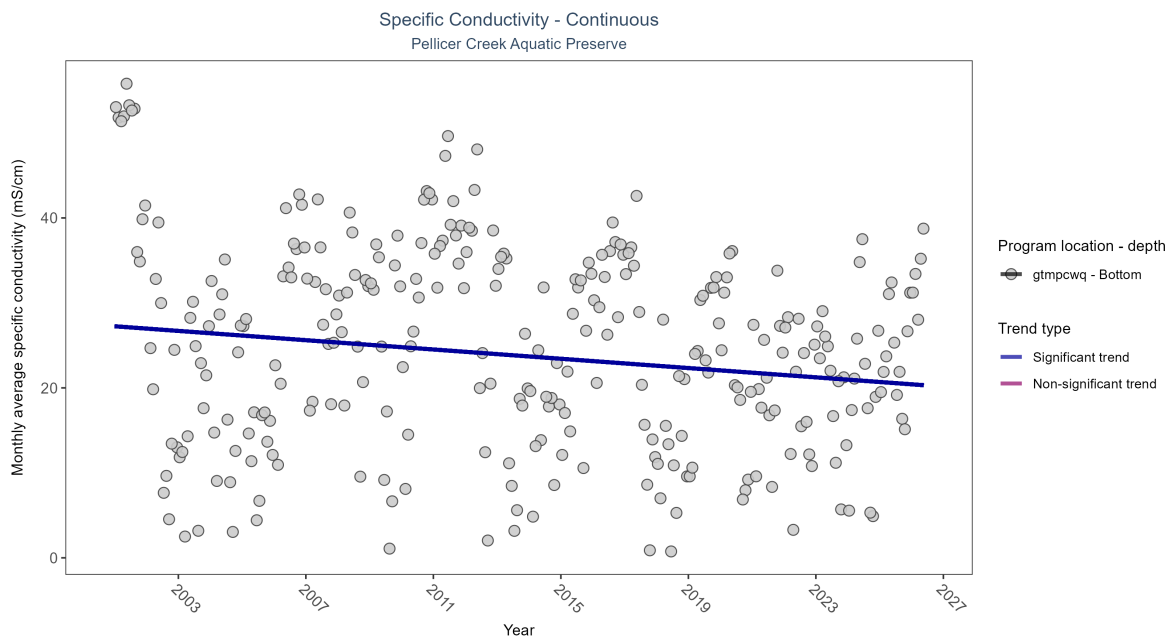


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At one program location, monthly average specific conductivity decreased by 0.27 mS/cm per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpcwq	Decreasing trend	756766	26	2001 - 2026	27.2	-0.12	27.25	-0.27	0

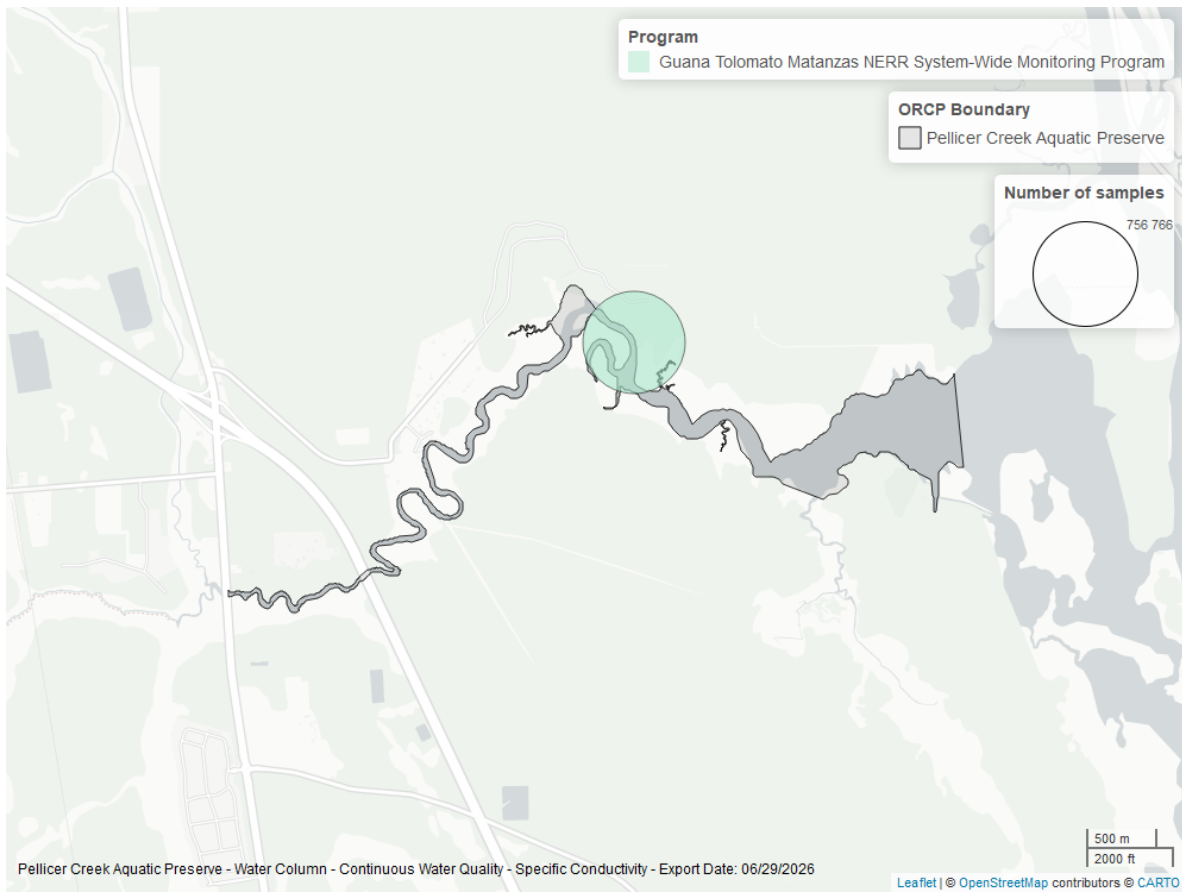


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

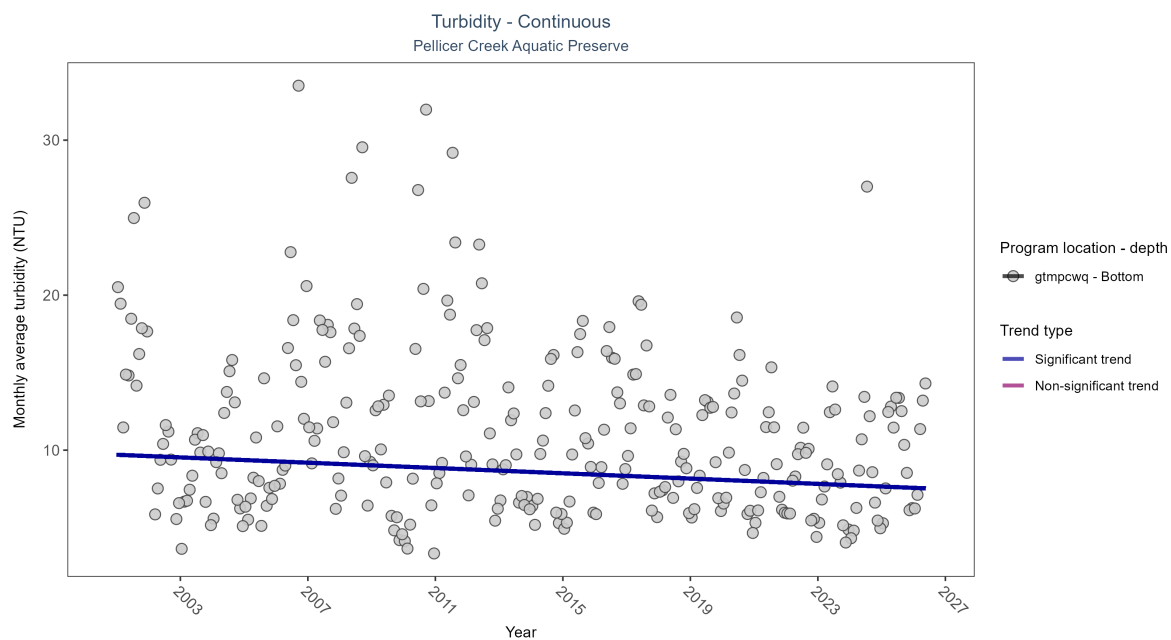


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: At one program location, monthly average turbidity decreased by 0.09 NTU per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpcwq	Decreasing trend	731428	26	2001 - 2026	9	-0.15	9.7	-0.09	0

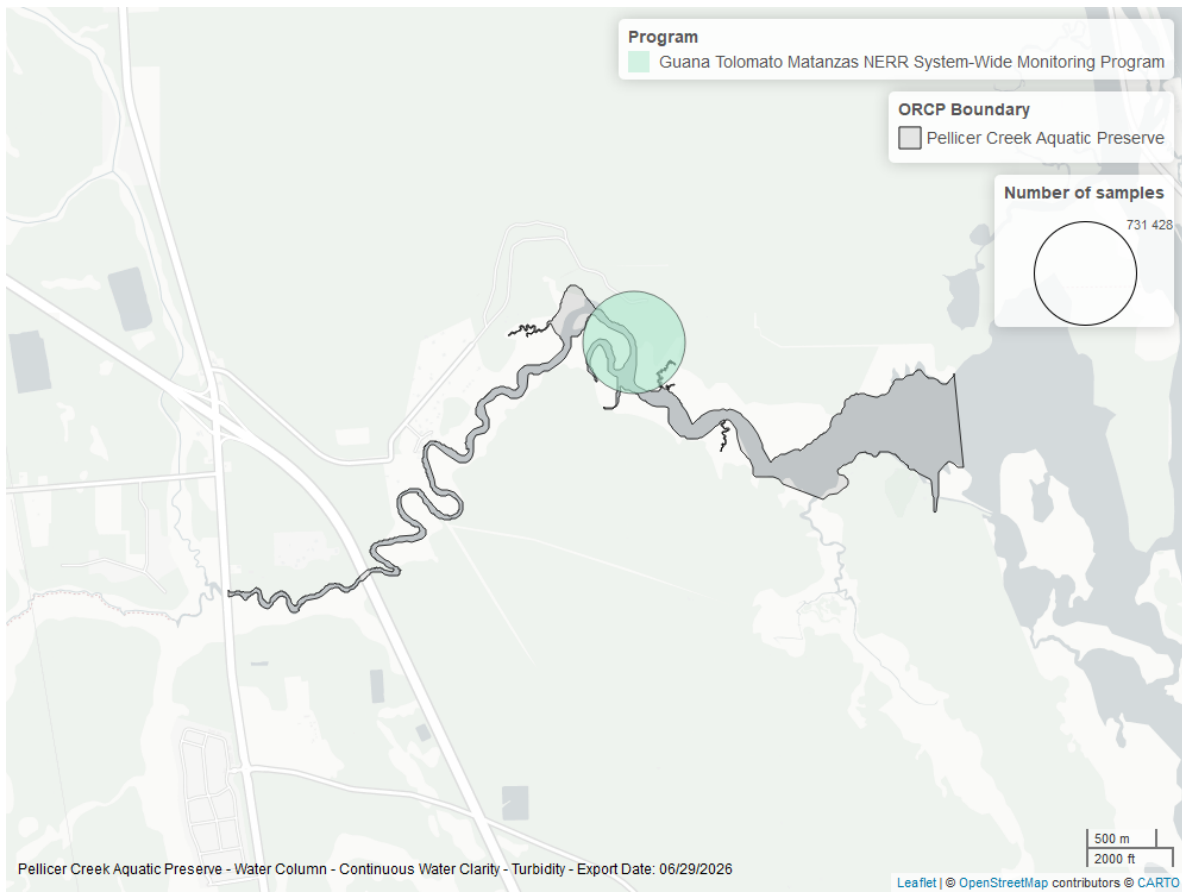


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

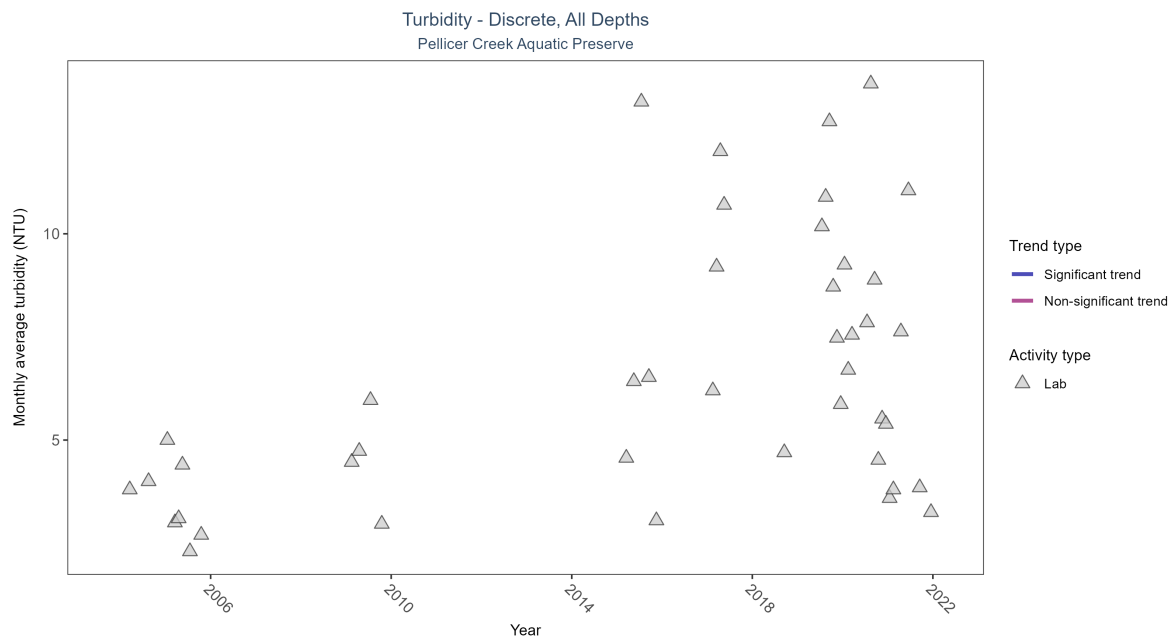


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: There was insufficient data to fit a model for turbidity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	237	9	2004 - 2021	7	-	-	-	-

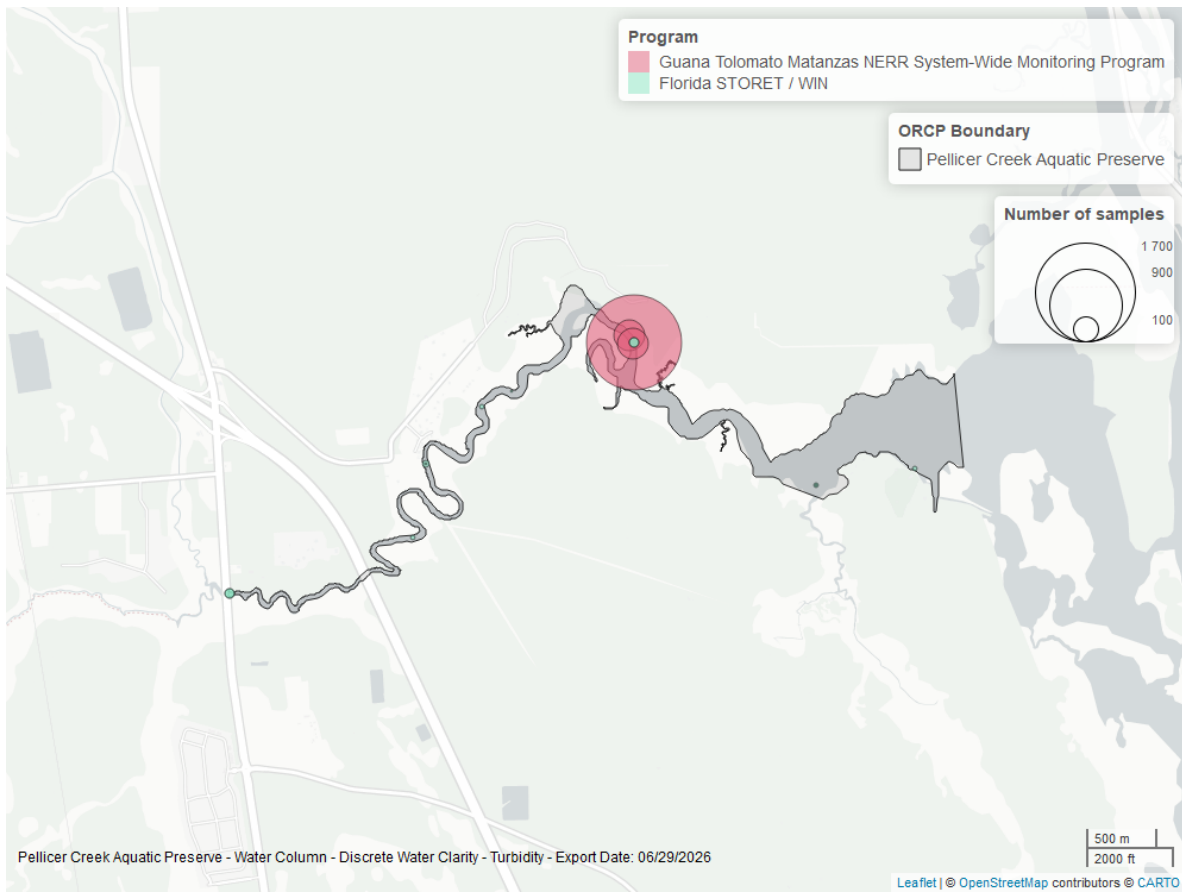


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

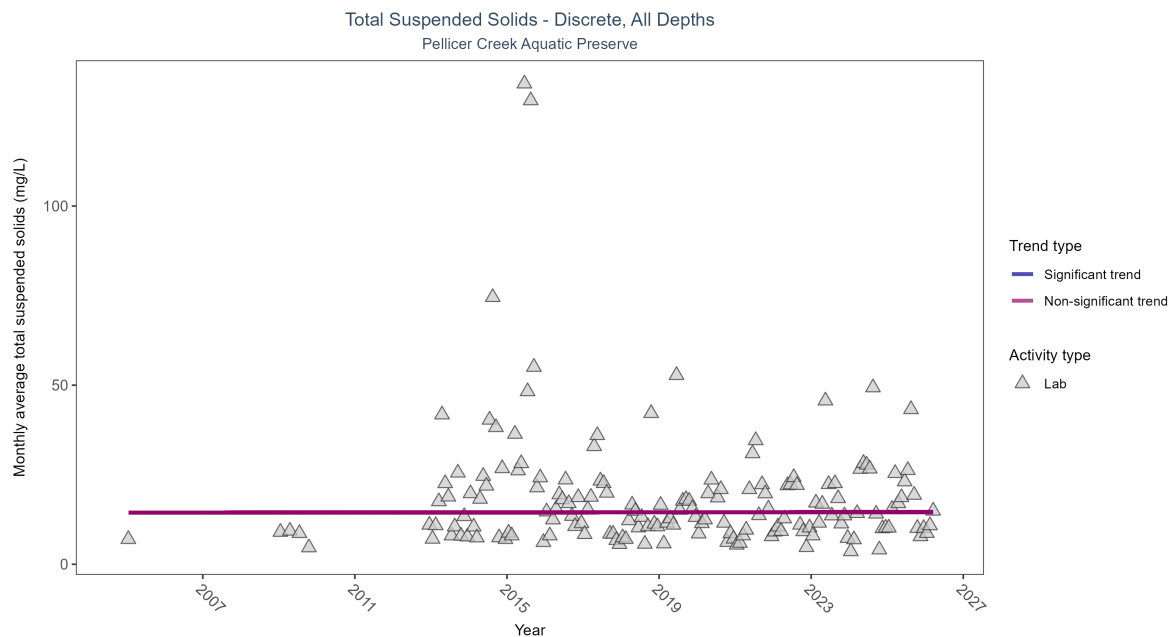


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Total suspended solids showed no detectable trend between 2005 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1985	17	2005 - 2026	14	0.01012	14.41506	0.00693	0.8836

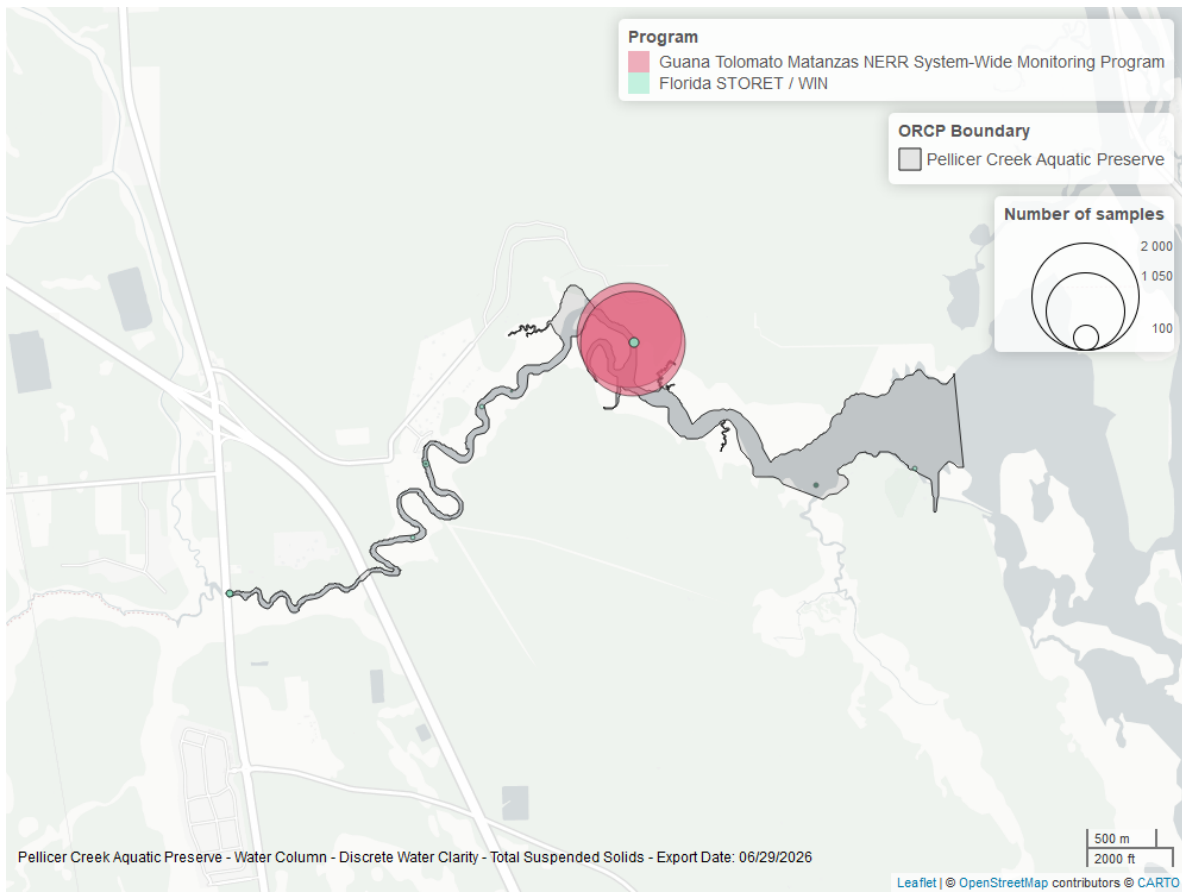


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

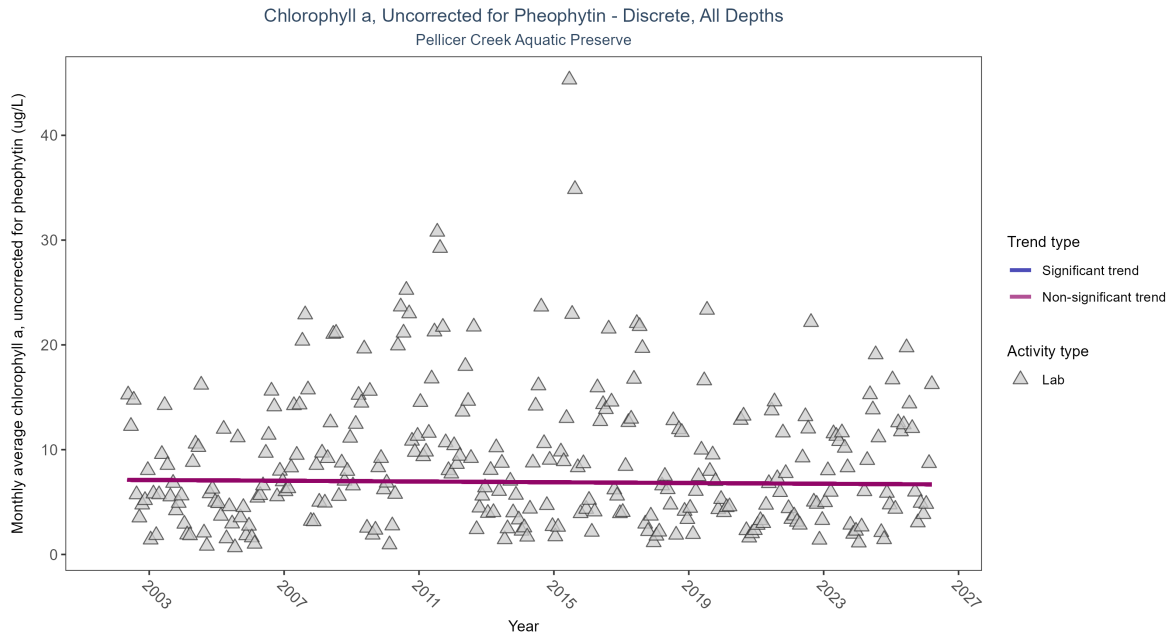


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	3751	25	2002 - 2026	6.9	-0.02682	7.123	-0.01795	0.5157

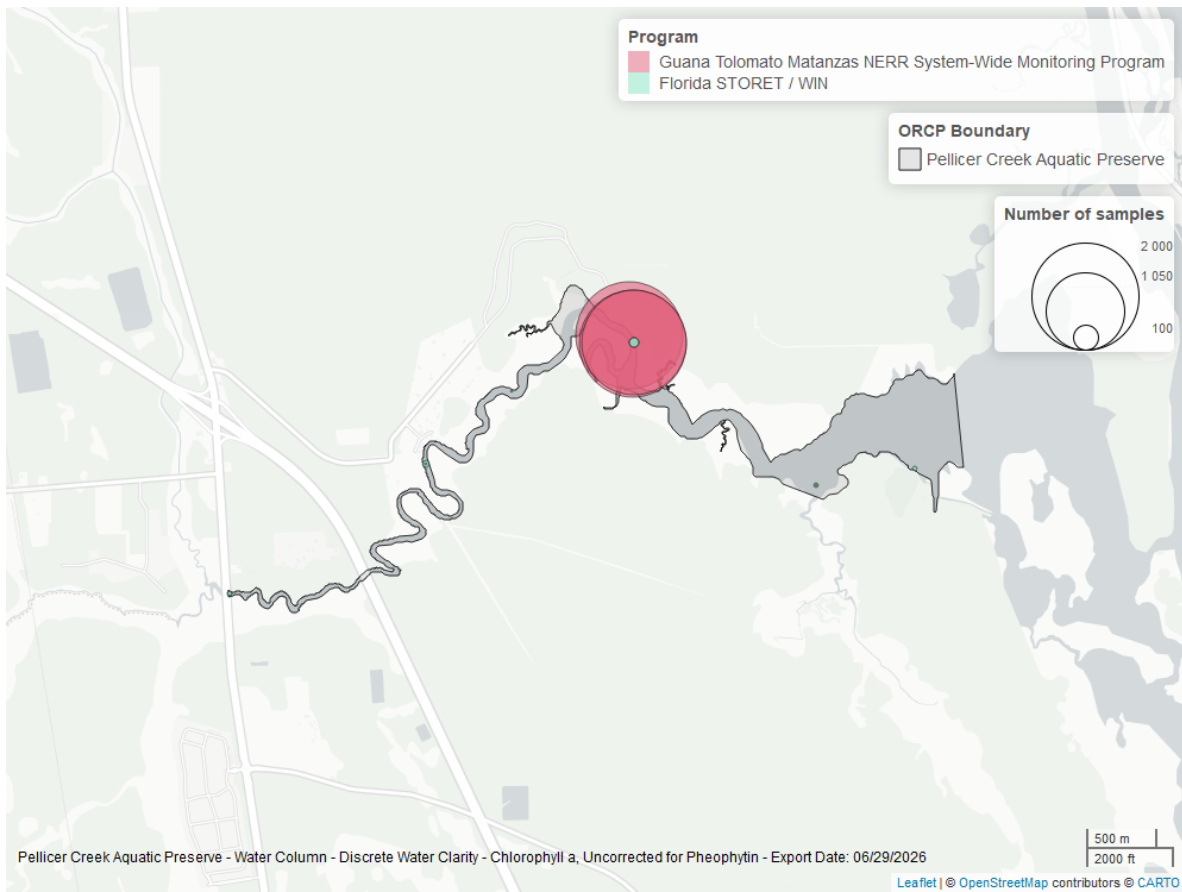


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

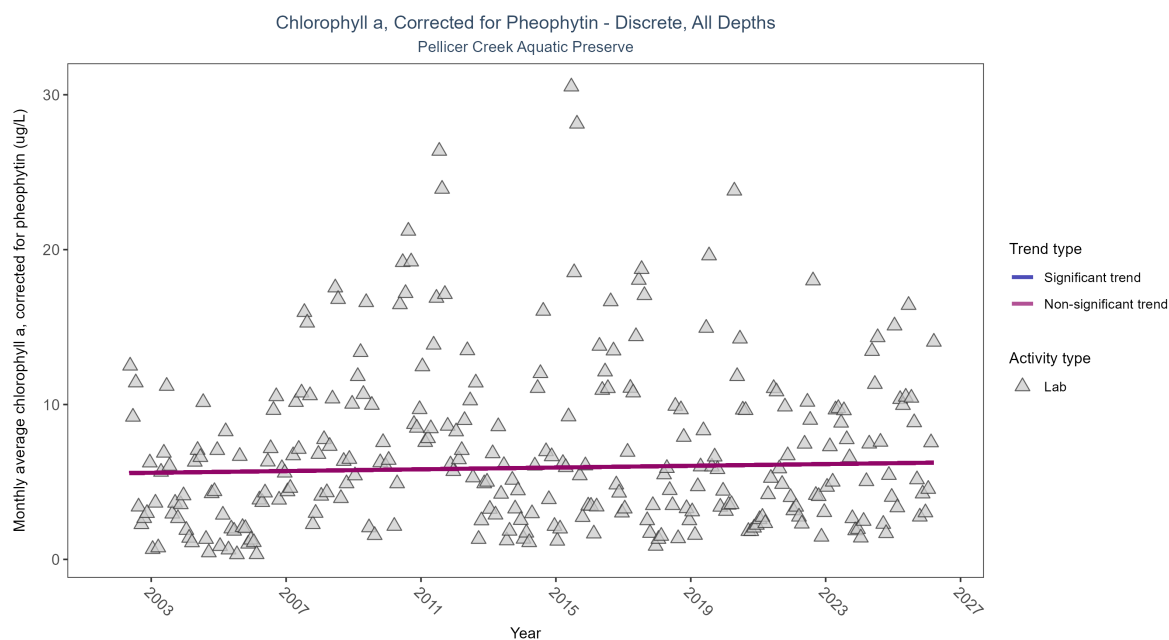


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	4671	25	2002 - 2026	5.2	0.04367	5.56394	0.02815	0.3089

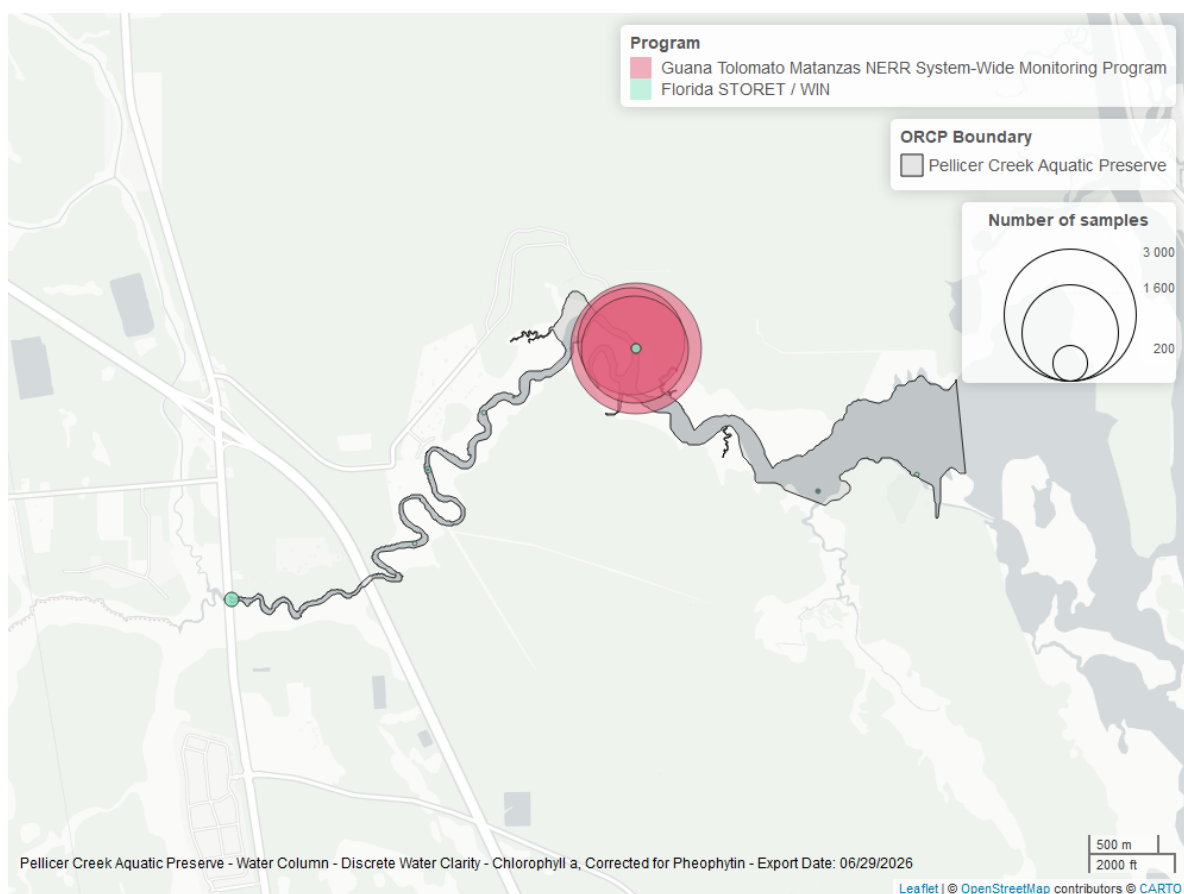


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

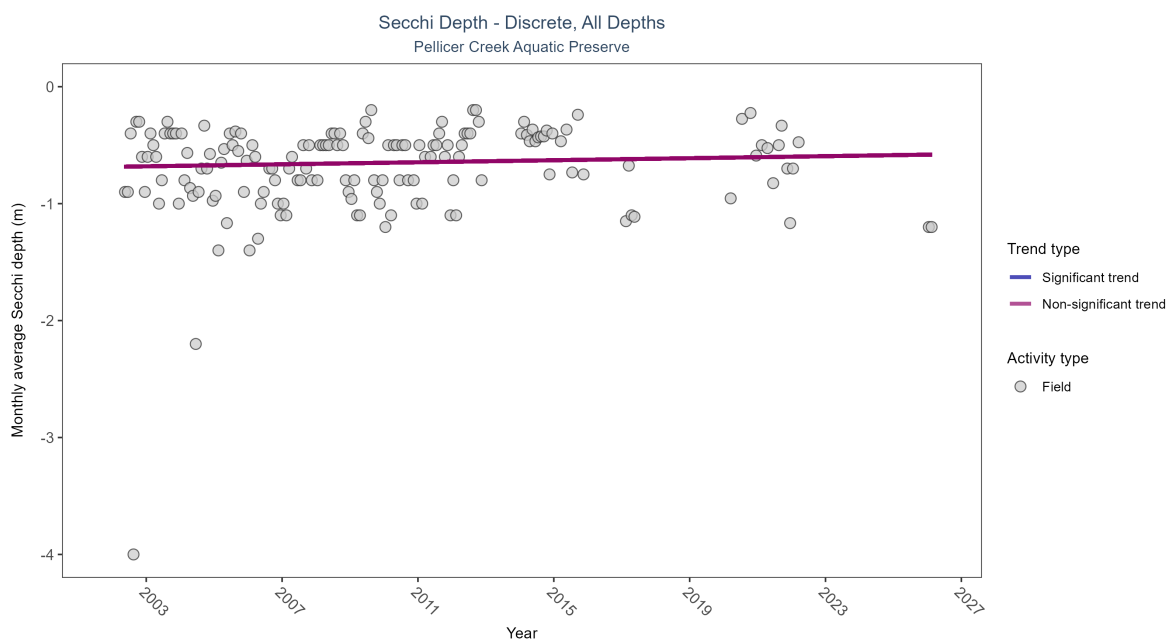


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Secchi depth showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	361	18	2002 - 2026	-0.6	0.10282	-0.68537	0.00434	0.1023

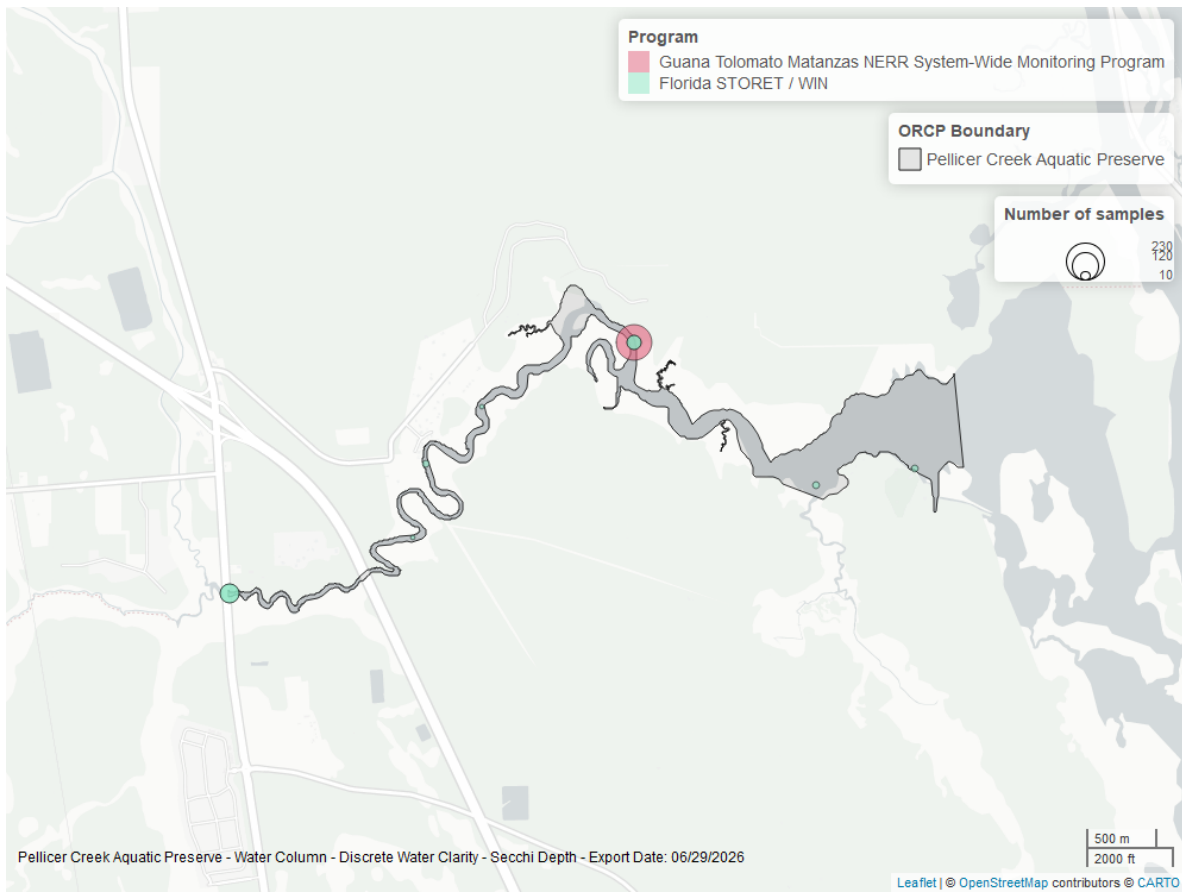


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

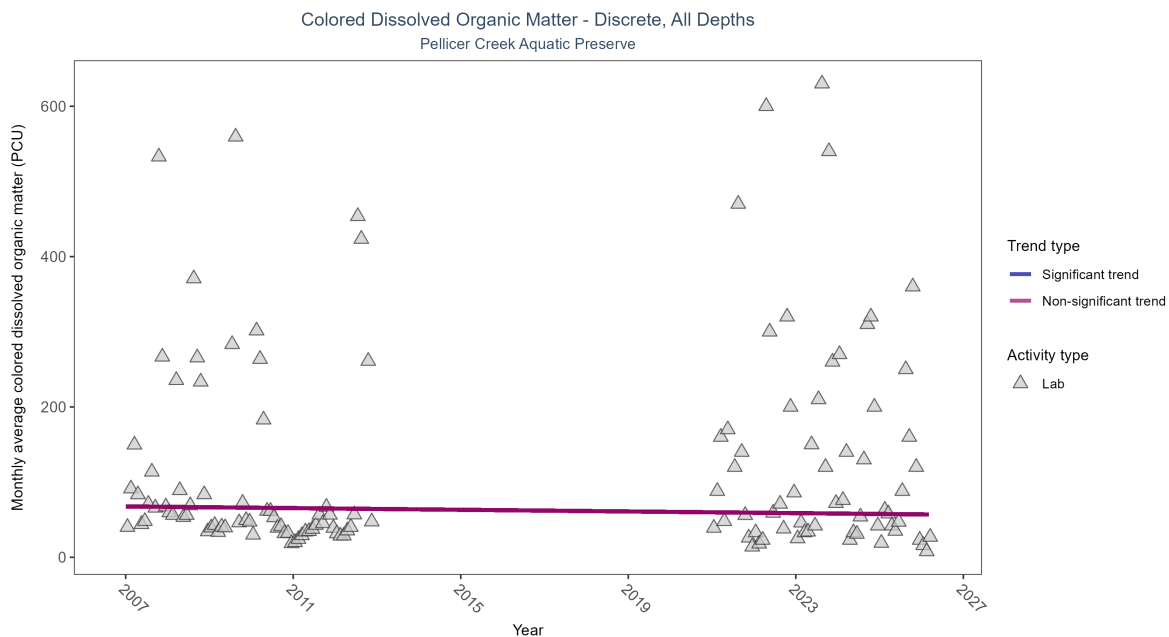


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Colored dissolved organic matter showed no detectable trend between 2007 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	969	12	2007 - 2026	54.6	-0.09467	67.67356	-0.55734	0.1401

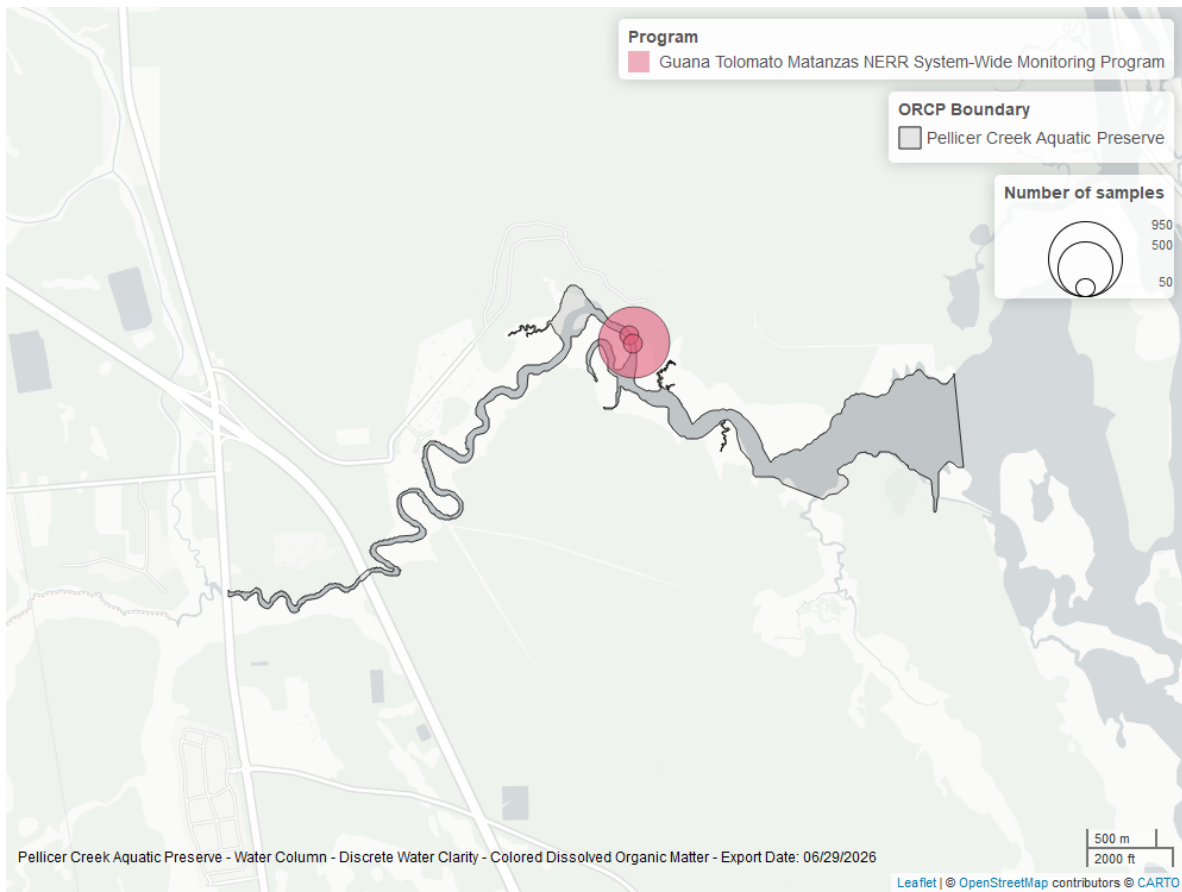


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Pellicer Creek Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.